

RESPONSIBLE JAVASCRIPT

JEREMY WAGNER — @MALCHATA — JEREMY.CODES

PRAIRIE DEV CON — WINNIPEG, MANITOBA, CANADA — JUNE 2019

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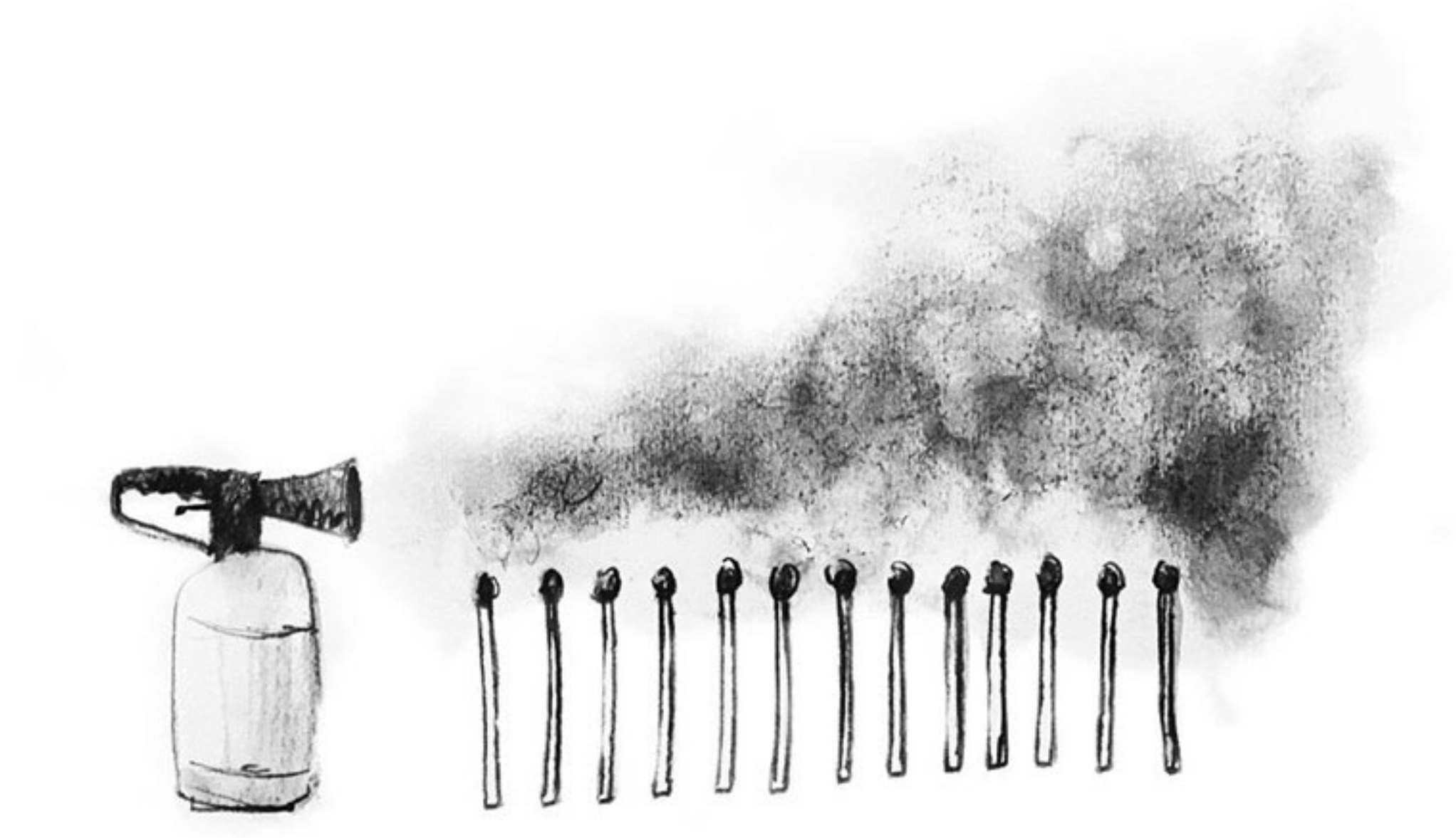
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A LIST APART



Responsible JavaScript: Part I

by [Jeremy Wagner](#) · March 28, 2019

Published in [Application Development](#), [JavaScript](#)

By the numbers, [JavaScript is a performance liability](#). If the trend persists, the median page will be shipping at least 400 KB of it before too long, and that's merely what's *transferred*. Like other text-based resources, JavaScript is almost always served compressed—but that might be the only thing we're getting consistently right in its delivery.

Unfortunately, while reducing resource transfer time is a big part of that whole performance thing, compression has no effect on how long browsers take to process a script once it arrives in its entirety. If a server sends 400 KB of compressed JavaScript, the actual amount browsers have to process after decompression is north of a megabyte. How well devices cope with these heavy workloads depends, well, on the *device*. [Much has been written](#) about how adept various devices are at processing lots of JavaScript, but the truth is, the amount of time it takes to process even a trivial amount of it varies greatly between devices.

Share this:



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Translations

SPHEXISHNESS

SPHEXISH

(of animal behavior) deterministic; preprogrammed

SPHEX

Sphex pensylvanicus





<code>npm install react</code>	<i>6.5 KB</i>
<code>npm install react-dom</code>	<i>103.7 KB</i>
<code>npm install react-router</code>	<i>21.6 KB</i>
<code>npm install react-redux</code>	<i>14.4 KB</i>



node_modules/

react-router@5.0.0

i

tree-shakeable

side-effect free

10 dependencies

npm

BUNDLE SIZE

21.6kB

MINIFIED

7.5kB

MINIFIED + GZIPPED

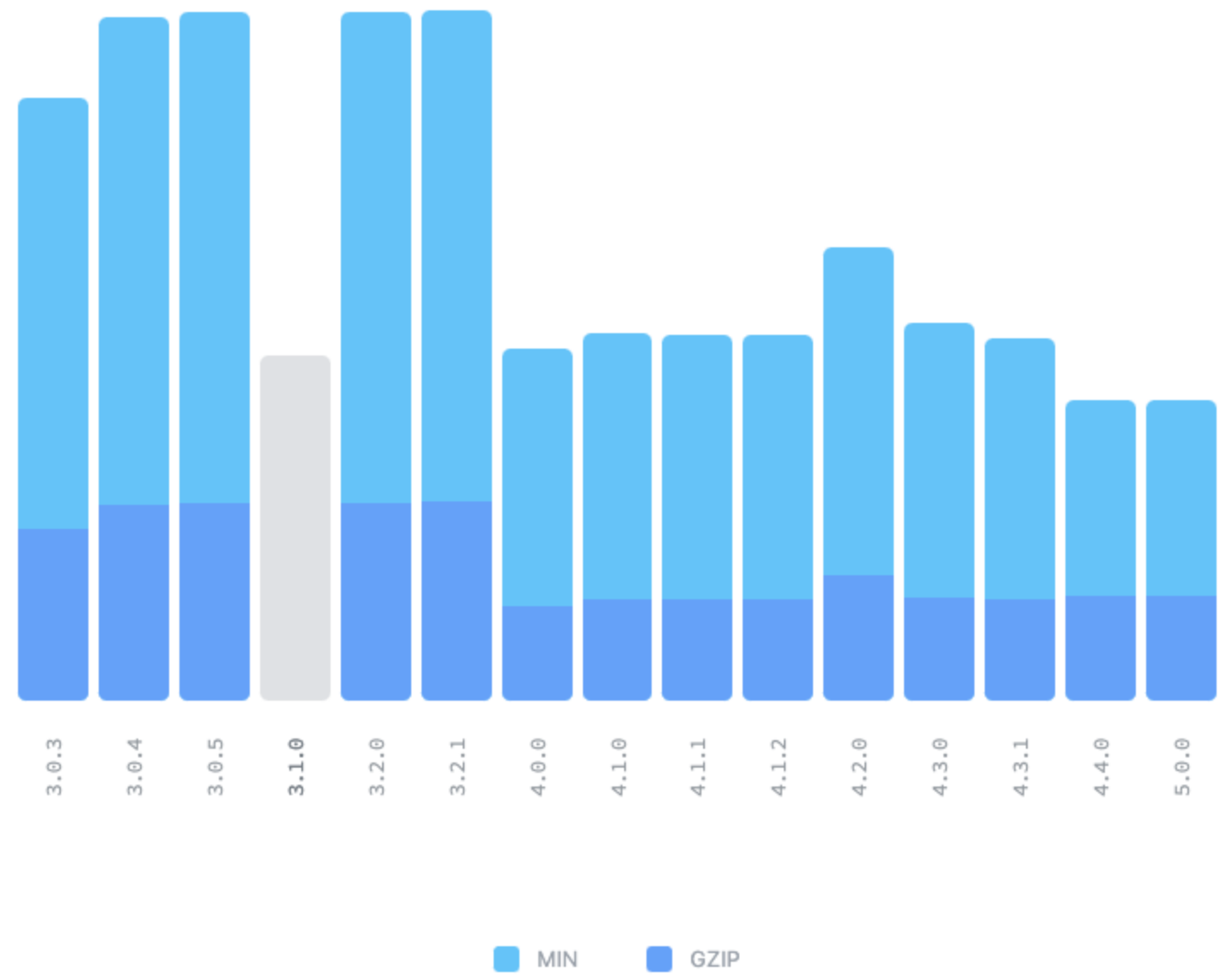
DOWNLOAD TIME

251ms

2G EDGE ¹

151ms

EMERGING 3G ¹



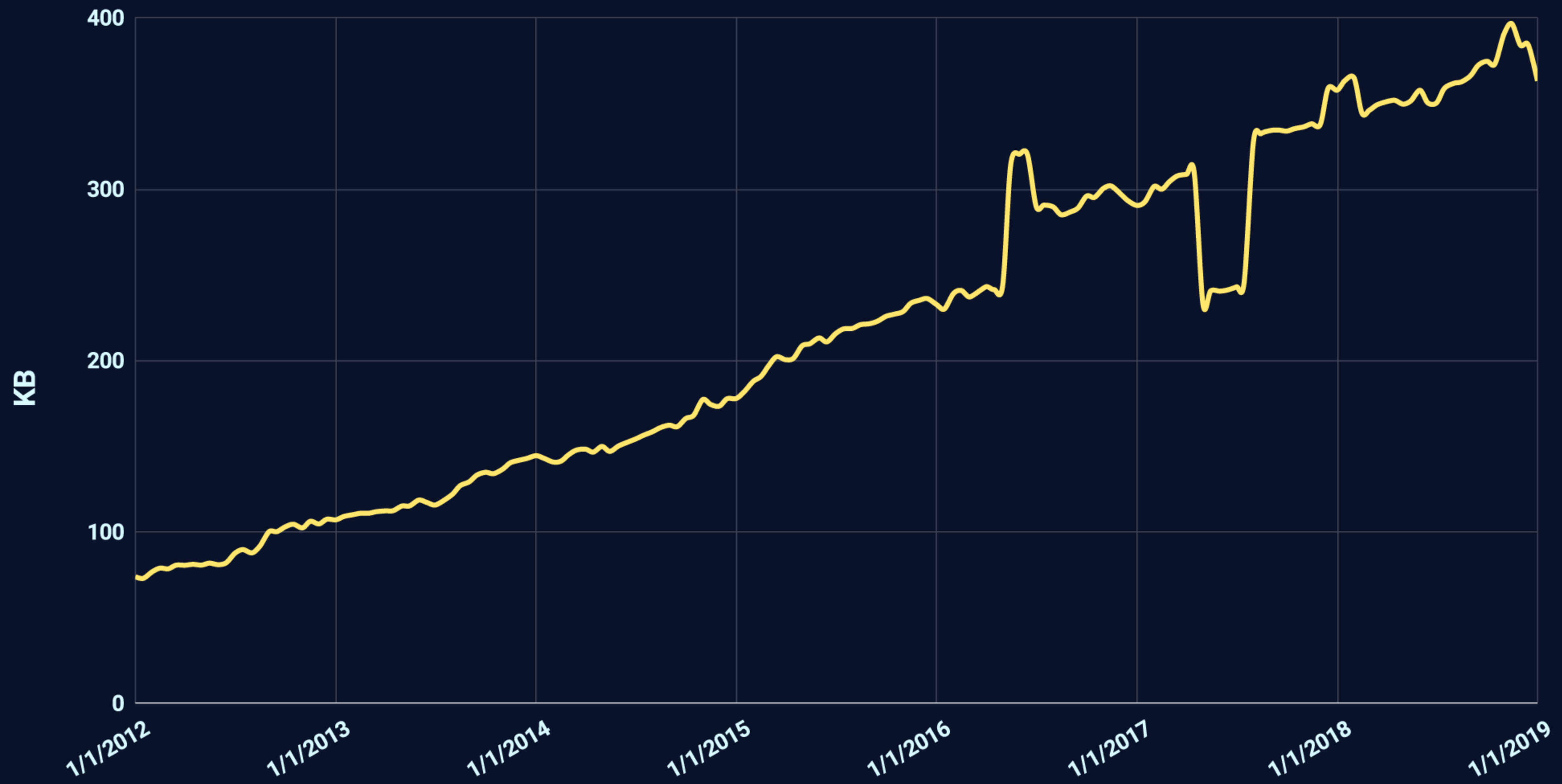
Composition



Note: These sizes represent the contribution made by dependencies (direct or transitive) to react-router's size. These may be different from the dependencies' standalone sizes.

JavaScript Delivered on Mobile Devices

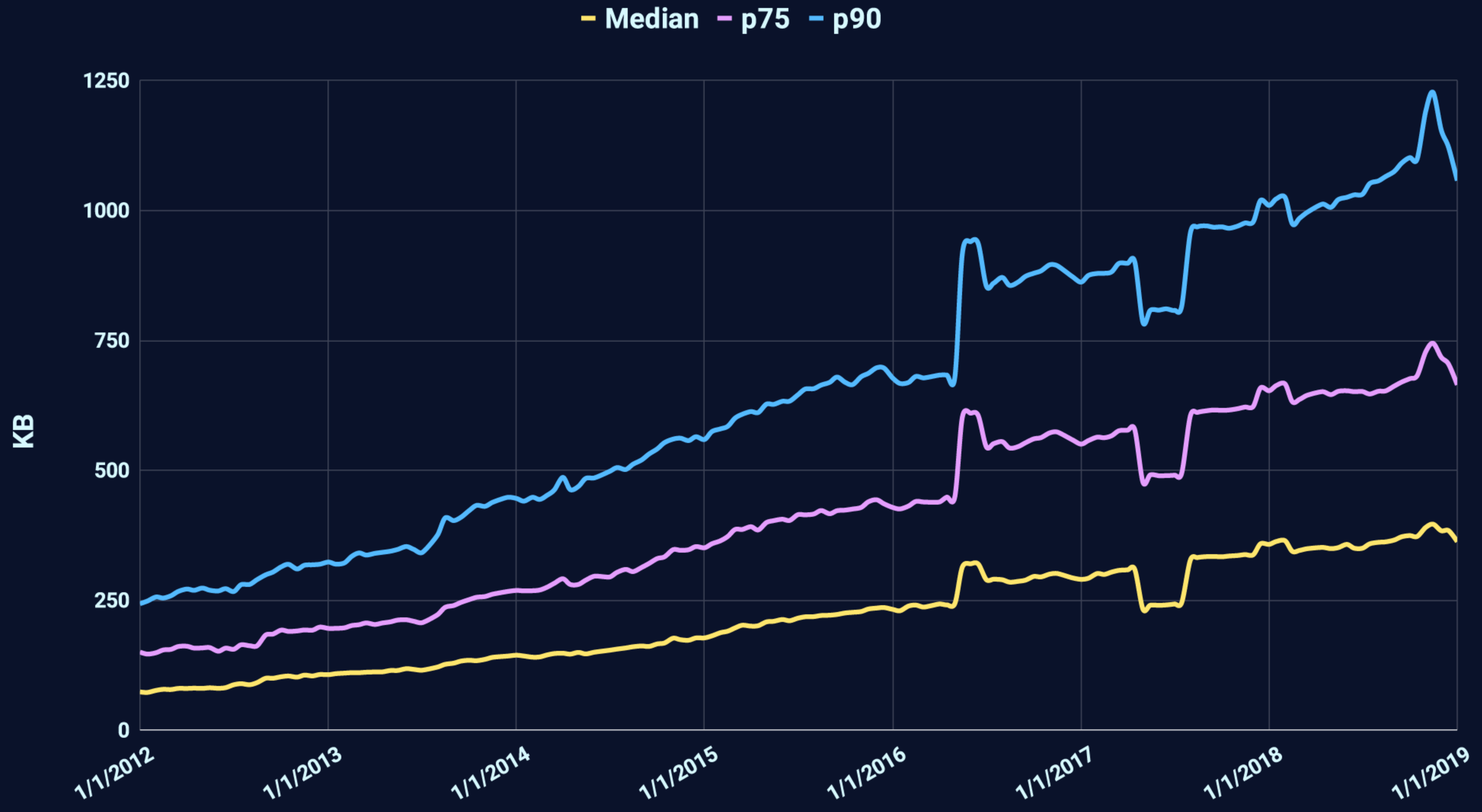
— Median



JavaScript Delivered on Mobile Devices



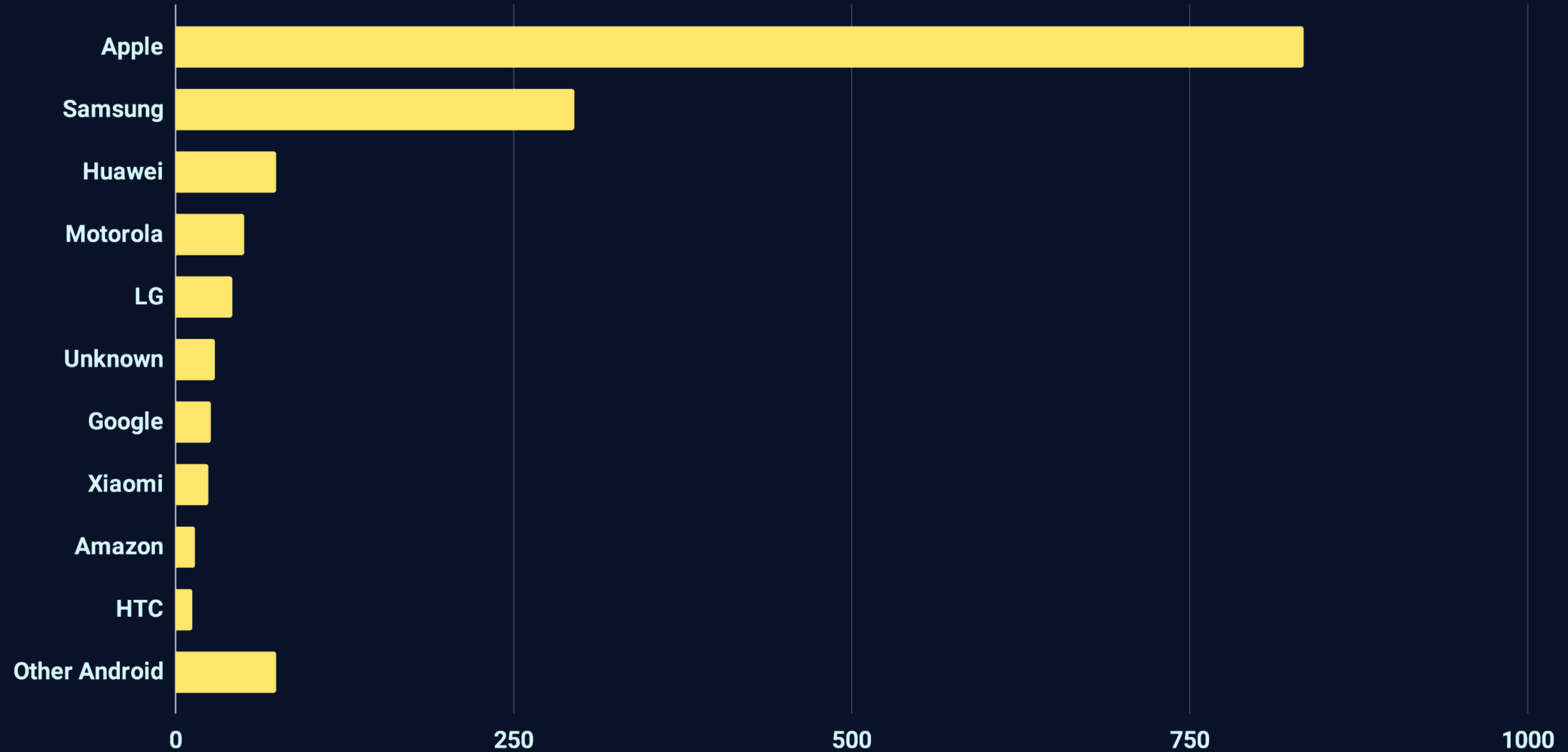
JavaScript Delivered on Mobile Devices

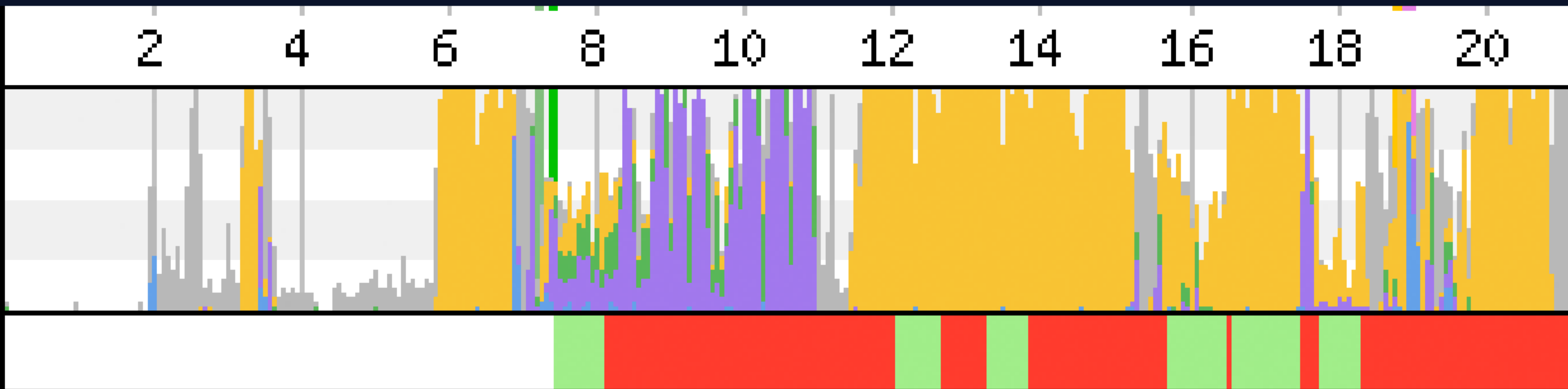






Mobile Devices on LegendaryTones.com Per Week





SCRIPT



RENDER



PAINT



NET



OTHER



SPHEXISHNESS

ANTI-SPHEXISHNESS

PAINT THE PICTURE

NOT THE FRAME

A LIST APART

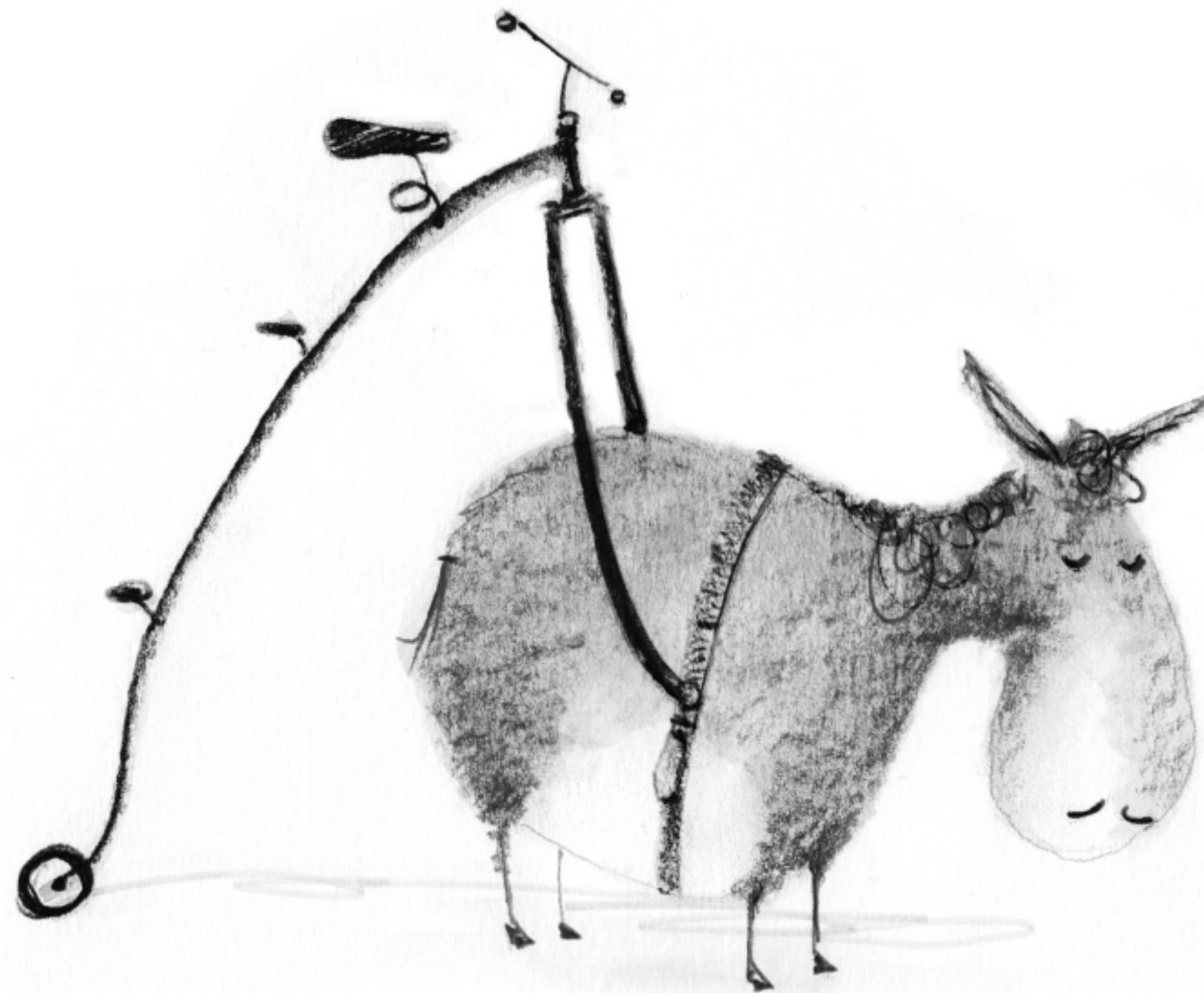


Illustration by [Dougal MacPherson](#)

Paint the Picture, Not the Frame: How Browsers Provide Everything Users Need

by [Eric Bailey](#) · February 07, 2019

Published in [Accessibility](#), [User Experience](#)

Kip Williams, professor of psychology sciences at Purdue University, conducted [a fascinating experiment](#) called “cyberball.” In his experiment, a test subject and two other participants played a computer game of catch. At a predetermined time, the test subject was excluded from the game, forcing them to only observe as the clock ran down.

However, the stakes for a quality design were pretty high: scrollbars are part of an application's interface, not a website's. In inclusive design, it's part of what we call *external consistency*.

External consistency is the idea that an object's functionality is informed and reinforced by similar implementations elsewhere. It's why you can flip a wall switch in most houses and be guaranteed the lights come on instead of flushing the toilet.



PRESS


```
import React, { Component } from "react";
import { validateEmail } from "helpers/validation";

class SignupForm extends Component {
  constructor (props) {
    super(props);

    this.handleSubmit = this.handleSubmit.bind(this);
    this.updateEmail = this.updateEmail.bind(this);
    this.state.email = "";
  }

  updateEmail (event) {
    this.setState({
      email: event.target.value
    });
  }

  handleSubmit () {
    // If the email checks out, submit
    if (validateEmail(this.state.email)) {
      // ...
    }
  }

  render () {
    return (
      <div>
        <span class="email-label">Enter your email:</span>
        <input type="text" id="email" onChange={this.updateEmail} />
        <div class="submit-button" onClick={this.handleSubmit}>Sign Up</div>
      </div>
    );
  }
}
```



```
render () {  
  return (  
    <div>  
      <span class="email-label">Enter your email:</span>  
      <input type="text" id="email" onChange={this.updateEmail} />  
      <div class="submit-button" onClick={this.handleSubmit}>Sign Up</div>  
    </div>  
  );  
}
```



```
render () {  
  return (  
    <form method="POST" action="/signup" onSubmit={handleSubmit}>  
      <label for="email" class="email-label">Enter your email:</label>  
      <input type="email" id="email" required />  
      <button>Sign Up</button>  
    </form>  
  );  
}
```



```
import React from "react";

const SignupForm = props => {
  const handleSubmit = event => {
    // Needed in case we're sending data to the server XHR-style
    // (but will still work if server-rendered with JS disabled).
    event.preventDefault();

    // Carry on...
  };

  return (
    <form method="POST" action="/signup" onSubmit={handleSubmit}>
      <label for="email" class="email-label">Enter your email:</label>
      <input type="email" id="email" required />
      <button>Sign Up</button>
    </form>
  );
};
```



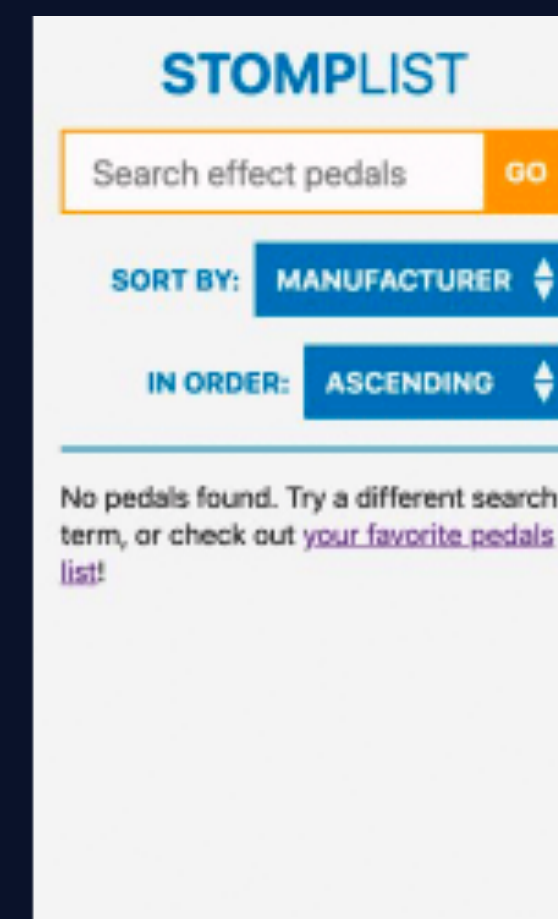

1 MS



2.07 S



5.24 S



CLIENT-SIDE RENDERING

1 MS

STOMPLIST

Search effect pedals

SORT BY:

IN ORDER:

No pedals found. Maybe try a different search term? Or perhaps browse [your favorite pedals list](#)?

2.07 S

STOMPLIST

Search effect pedals

SORT BY:

IN ORDER:

No pedals found. Maybe try a different search term? Or perhaps browse [your favorite pedals list](#)?

5.24 S

STOMPLIST

Search effect pedals

SORT BY:

IN ORDER:

No pedals found. Maybe try a different search term? Or perhaps browse [your favorite pedals list](#)?

SERVER-SIDE RENDERING

(WITH CLIENT-SIDE HYDRATION)


```
<link rel="prefetch" href="/products/snes-console">
```




npm v1.0.0 gzip size 782 B downloads 37k build passing

quicklink

Faster subsequent page-loads by prefetching in-viewport links during idle time

How it works

Quicklink attempts to make navigations to subsequent pages load faster. It:

- **Detects links within the viewport** (using [Intersection Observer](#))
- **Waits until the browser is idle** (using [requestIdleCallback](#))
- **Checks if the user isn't on a slow connection** (using `navigator.connection.effectiveType`) or has data-saver enabled (using `navigator.connection.saveData`)
- **Prefetches URLs to the links** (using `<link rel=prefetch>` or XHR). Provides some control over the request priority (can switch to `fetch()` if supported).

dnstradamus



(graphic by [@flipty](#))

ES5 (.js) version

Uncompressed	1.16 kB	gzip	663 B	brotli	567 B
--------------	---------	------	-------	--------	-------

ES6 (.mjs) version

Uncompressed	1.03 kB	gzip	639 B	brotli	555 B
--------------	---------	------	-------	--------	-------




```
<!-- Native lazy loading in action -->  

```



```
<!-- Hybrid lazy loading -->  

```



```
if ("loading" in HTMLImageElement.prototype) {
    document.querySelectorAll("img[loading=lazy]").forEach(imageEl => {
        imageEl.src = imageEl.dataset.src;
        imageEl.srcset = imageEl.dataset.srcset;
    });
} else {
    let scriptEl = document.createElement("script");
    scriptEl.src = "/js/lazyload.js";
    document.body.appendChild(scriptEl);
}
```




ABOUT THE AUTHOR

Andrea has been programming computers for 30 years and he's now a front-end development addicted who loves to craft websites with the user experience at the ... [More about Andrea...](#)

MAY 6, 2019 • [6 comments](#)

Hybrid Lazy Loading: A Progressive Migration To Native Lazy Loading

QUICK SUMMARY ➡ Native lazy loading is coming to the web. Since it doesn't depend on JavaScript, it will revolutionize the way we lazy load content today, making it easier for developers to lazy load images and iframes. But it's not a feature we can polyfill, and it will take some time before it becomes usable across all browsers. In this article, you'll learn how it works and how you can progressively replace your JavaScript-driven lazy loading with its native alternative, thanks to hybrid lazy loading.

IN THE PAST FEW WEEKS, YOU MIGHT HAVE HEARD OR READ ABOUT native lazy loading, which is coming to Chromium 75 in the upcoming months.

"Yeah, great news, but we'll have to wait until all browsers support it."

 12 min read

 [JavaScript](#), [Performance](#), [Optimization](#)

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BROWSERS GIVE US
A LOT OF FREE STUFF

THE TOOLS ARE
NOT INFALLIBLE



PHOTO CREDIT: JOHN HOEY



Jason Miller



@_developit

Transpile less

12:20 PM - 25 Mar 2019


```
// Untransformed code:  
function logger(message, level = "log") {  
    console[level](message);  
}
```



```
// Babel-transformed code:
```

```
function logger(message) {  
  var level = arguments.length > 1 && arguments[1] !== undefined ? arguments[1] : "log";  
  console[level](message);  
}
```



```
// Code that Babel won't touch:  
function logger(message, level) {  
  console[level || "log"](message);  
}
```



```
export class User {  
  constructor (id, name, email) {  
    this.id = id;  
    this.name = name;  
    this.email = email;  
  }  
  
  getId () {  
    return this.id;  
  }  
  
  getName () {  
    return this.name;  
  }  
  
  getEmail () {  
    return this.email;  
  }  
}
```



```
"use strict";

Object.defineProperty(exports, "__esModule", {
  value: true
});
exports.User = void 0;

function _classCallCheck(instance, Constructor) { if (!(instance instanceof Constructor)) { throw new TypeError("Cannot call a class as a function"); } }

function _defineProperties(target, props) { for (var i = 0; i < props.length; i++) { var descriptor = props[i]; descriptor.enumerable = descriptor.enumerable || false; descriptor.configurable = true; if ("value" in descriptor) descriptor.writable = true; Object.defineProperty(target, descriptor.key, descriptor); } }

function _createClass(Constructor, protoProps, staticProps) { if (protoProps) _defineProperties(Constructor.prototype, protoProps); if (staticProps) _defineProperties(Constructor, staticProps); return Constructor; }

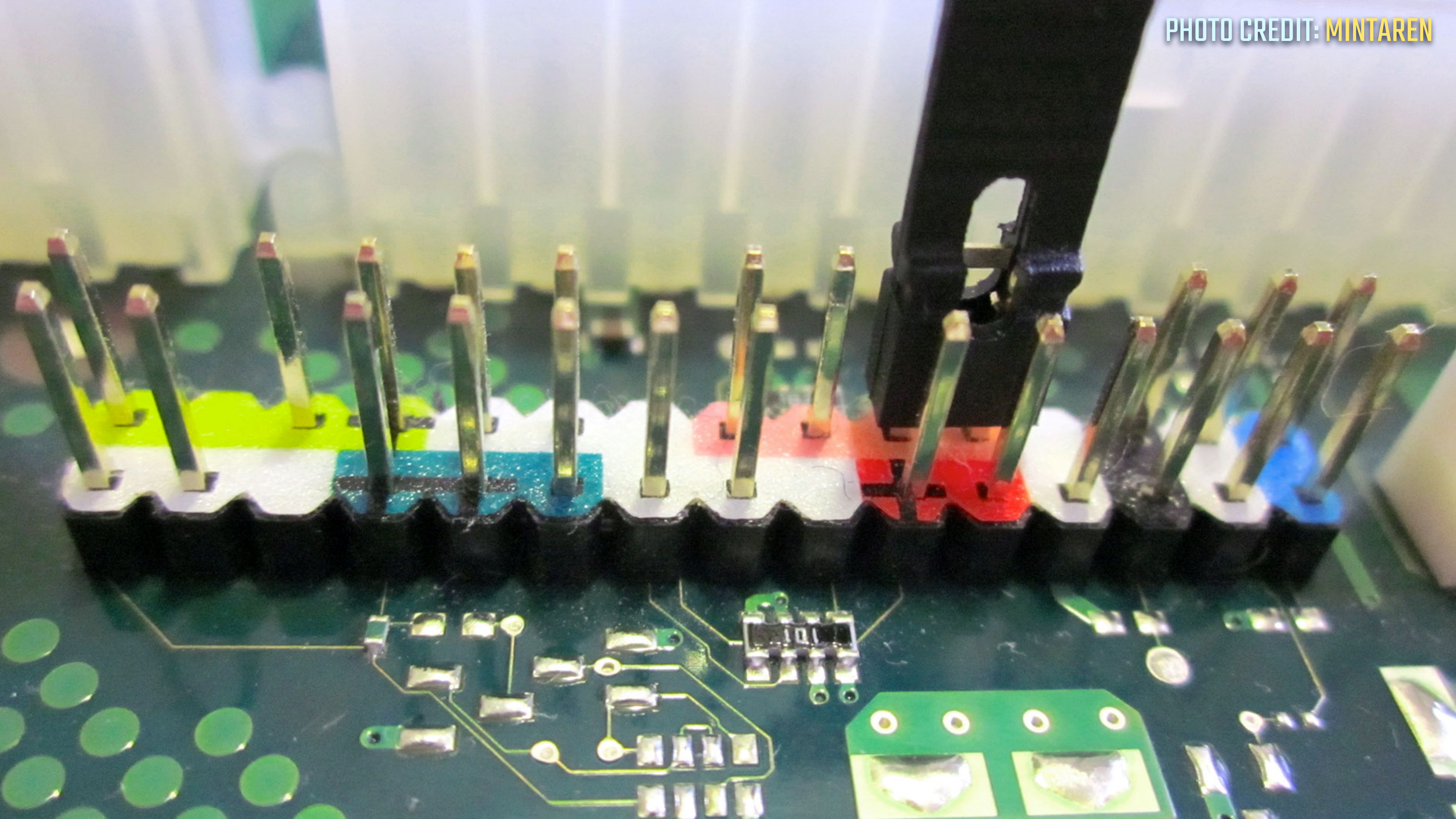
var User =
/*#__PURE__*/
function () {
  function User(id, name, email) {
    _classCallCheck(this, User);

    this.id = id;
    this.name = name;
    this.email = email;
  }

  _createClass(User, [{
    key: "getId",
    value: function getId() {
      return this.id;
    }
  }, {
    key: "getName",
    value: function getName() {
      return this.name;
    }
  }, {
    key: "getEmail",
    value: function getEmail() {
      return this.email;
    }
  }]);

  return User;
}();

exports.User = User;
```


```
presets: [  
  [  
    "@babel/preset-env", {  
      modules: false,  
      useBuiltIns: "entry",  
      corejs: 3,  
      targets: "> 0.25%, IE > 10, Firefox ESR, not dead"  
    }  
  ]  
]
```

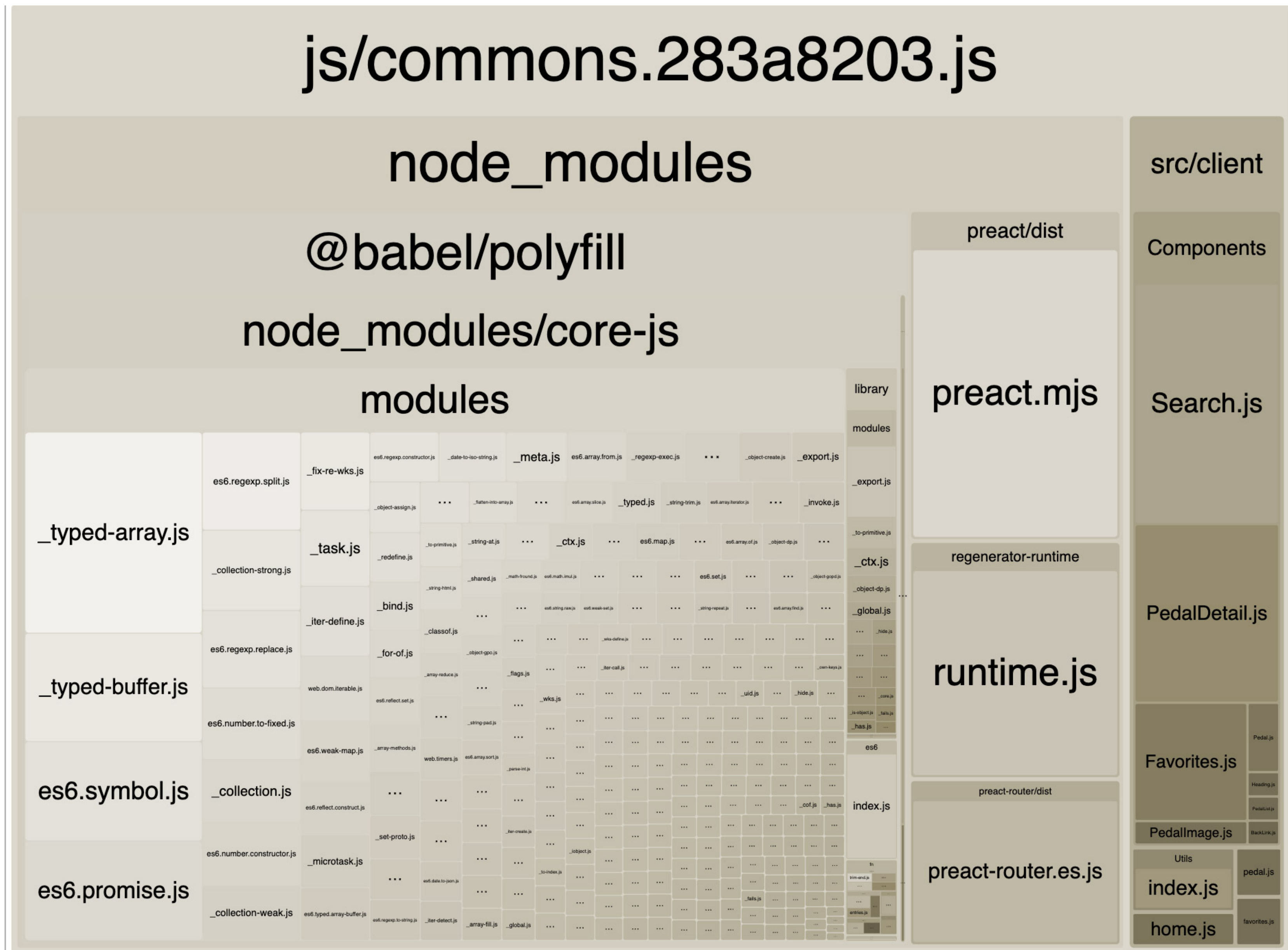




Gzipped

X

- ✓ All (**116.59 KB**)
- ✓ js/commons.283a8203.js (**112.17 KB**)
- ✓ js/favorites.a8b9ec0f.js (**1.48 KB**)
- ✓ js/pedal.ac9b429d.js (**1.47 KB**)
- ✓ js/home.12464d05.js (**1.47 KB**)



```
useBuiltIns: "entry"
```



```
presets: [  
  [  
    "@babel/preset-env", {  
      modules: false,  
      useBuiltIns: "usage",  
      corejs: 3,  
      targets: "> 0.25%, IE > 10, Firefox ESR, not dead"  
    }  
  ]  
]
```

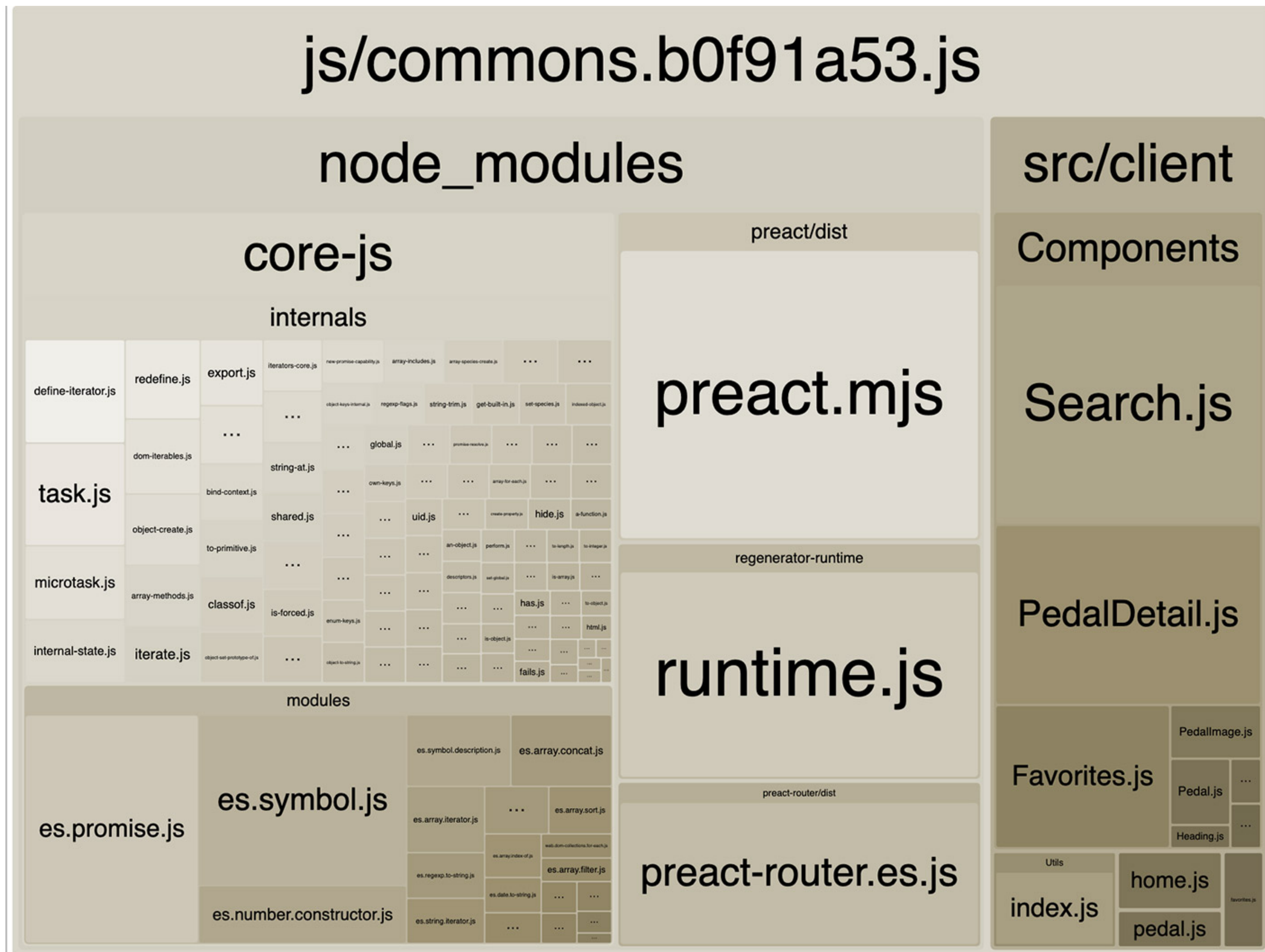

Stat

Parsed

Gzipped

Enter regexp

X



```
useBuiltIns: "usage"
```



```
presets: [  
  [  
    "@babel/preset-env", {  
      modules: false,  
      useBuiltIns: "usage",  
      corejs: 3,  
      targets: "> 0.25%, IE > 10, Firefox ESR, not dead"  
    }  
  ]  
]
```



```
presets: [  
  [  
    "@babel/preset-env", {  
      modules: false,  
      useBuiltIns: "usage",  
      loose: true,  
      corejs: 3,  
      targets: "> 0.25%, IE > 10, Firefox ESR, not dead"  
    }  
  ]  
]
```


Treemap sizes:

Stat

Parsed

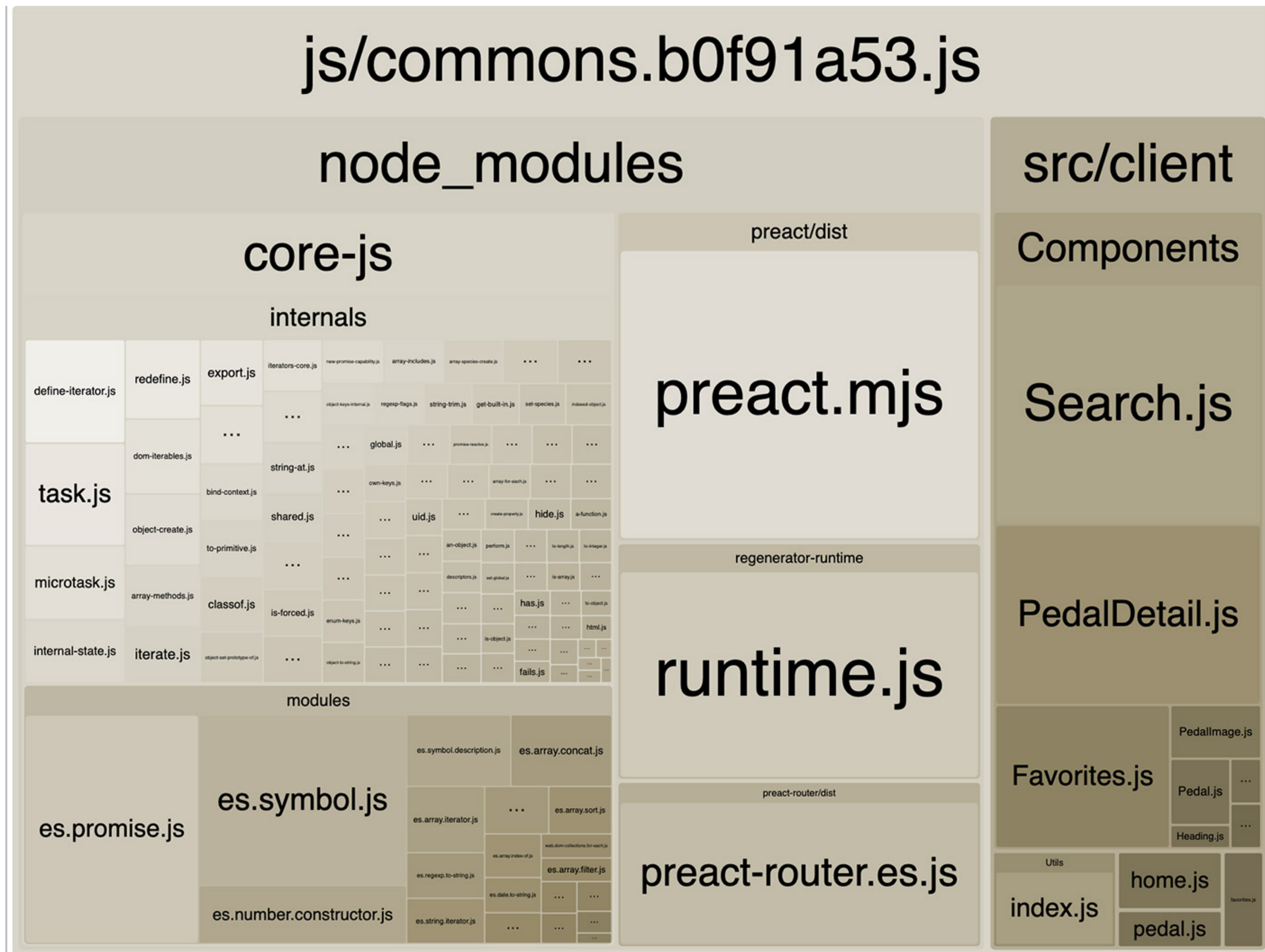
Gzipped

Search modules:

X

Show chunks:

- ✓ All (**68.48 KB**)
- ✓ js/commons.b0f91a53.js (**64.05 KB**)
- ✓ js/favorites.760ee393.js (**1.48 KB**)
- ✓ js/pedal.2f0b52ea.js (**1.47 KB**)
- ✓ js/home.9c5b675b.js (**1.47 KB**)



loose: false

Treemap sizes:

Stat

Parsed

Gzipped

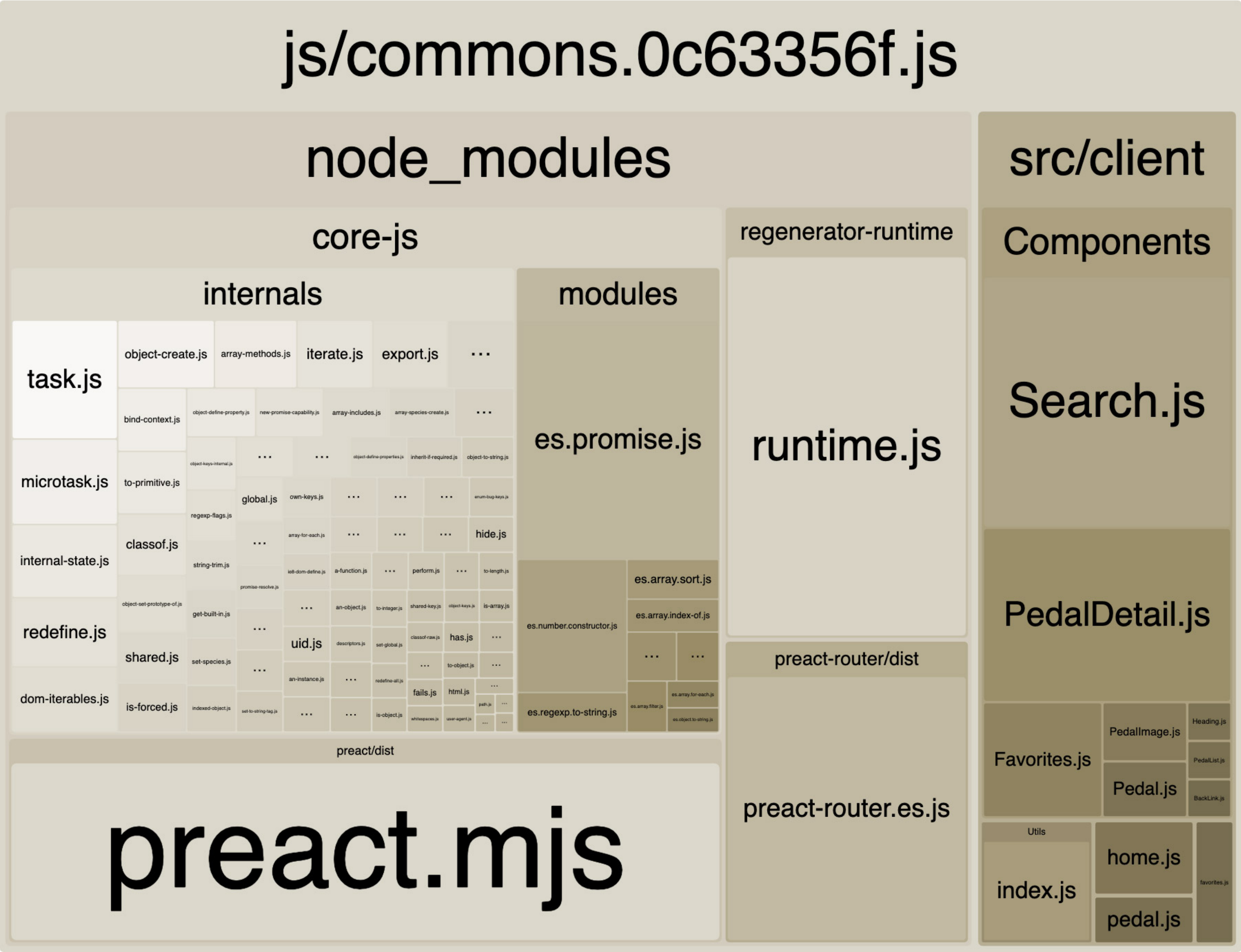
Search modules:

Enter regexp

X

Show chunks:

- ☒ All (56.09 KB)
- ☒ js/commons.0c63356f.js (51.67 KB)
- ☒ js/favorites.6eb32550.js (1.48 KB)
- ☒ js/pedal.7c773d3d.js (1.47 KB)
- ☒ js/home.5331f7ef.js (1.47 KB)



loose: true

2ality – JavaScript and more

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Babel 6: loose mode

[2015-12-12] [dev](#), [javascript](#), [jstools](#), [babel](#), [esnext](#)

Babel's loose mode transpiles ES6 code to ES5 code that is less faithful to ES6 semantics. This blog post explains how that works and what the pros and cons are (spoiler: normally not recommended).

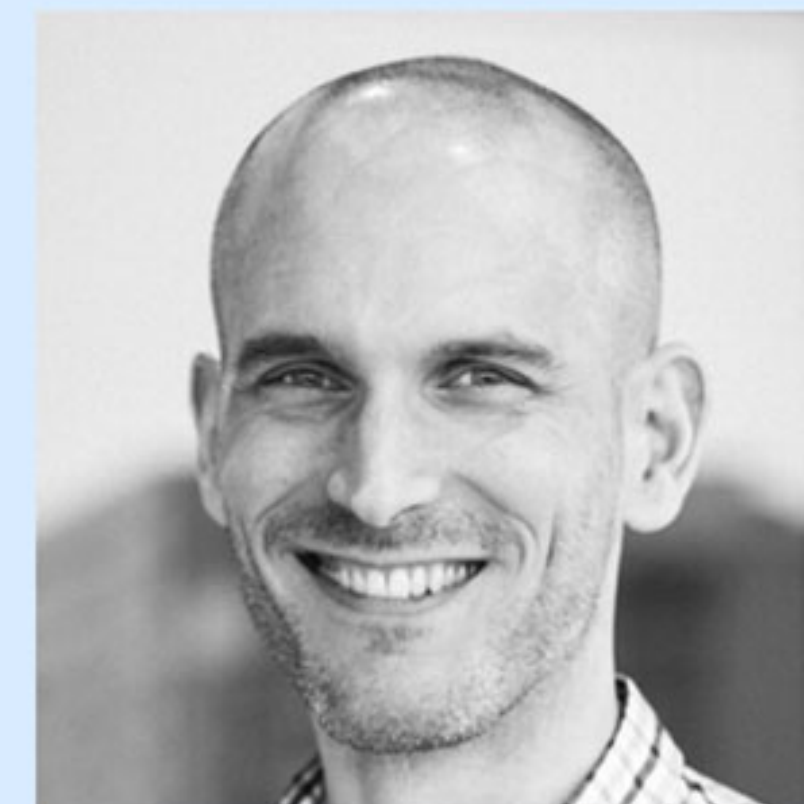
Starting point for [this series of posts on Babel 6: "Configuring Babel 6"](#) [explains the basics: configuration files, presets, plugins, etc.]

1. Two modes

Many plugins in Babel have two modes:

- A normal mode follows the semantics of ECMAScript 6 as closely as possible.
- A loose mode produces simpler ES5 code.

Normally, it is recommended to not use loose mode. The pros and cons are:



Dr. Axel Rauschmayer
[Homepage](#) | [Twitter](#)



DIFFERENTIAL SERVING


```
<!-- The way we've always done it: -->  
<script defer src="/js/app.js"></script>
```



```
<!-- Modern browsers get this: -->  
<script type="module" src="/js/app.mjs"></script>  
<!-- Legacy browsers get this: -->  
<script nomodule defer src="/js/app.js"></script>
```



```
// Config for legacy browsers
presets: [
  [
    "@babel/preset-env", {
      modules: false,
      useBuiltIns: "usage",
      targets: "> 0.25%, IE > 10, Firefox ESR"
    }
  ]
]
```



```
// Config for modern browsers
presets: [
  [
    "@babel/preset-env", {
      modules: false,
      targets: {
        esmodules: true
      }
    }
  ]
]
```



```
// babel.config.js
module.exports = {
  env: {
    clientLegacy: {
      presets: [
        [
          "@babel/preset-env", {
            modules: false,
            targets: "> 0.25%, IE > 10, Firefox ESR"
          }
        ]
      ]
    },
    clientModern: {
      presets: [
        [
          "@babel/preset-env", {
            modules: false,
            targets: {
              esmodules: true
            }
          }
        ]
      ]
    }
  }
};
```

```
// Legacy config ...
module: {
  rules: [
    {
      test: /\.m?js$/i,
      exclude: /node_modules/i,
      use: [
        {
          loader: "babel-loader",
          options: {
            envName: "clientLegacy"
          }
        }
      ]
    }
  ]
},
// ...
```



```
// Modern config ...
module: {
  rules: [
    {
      test: /\.m?js$/i,
      exclude: /node_modules/i,
      use: [
        {
          loader: "babel-loader",
          options: {
            envName: "clientModern"
          }
        }
      ]
    }
  ]
},
// ...
```

 localhost	200	1.3 KB	576 ms						
 styles.css	200	2.3 KB	580 ms						
 commons.70ac62b7.mjs	200	7.9 KB	614 ms						
 home.1350bbcf.mjs	200	1.1 KB	576 ms						

Treemap sizes:

Stat

Parsed

Gzipped

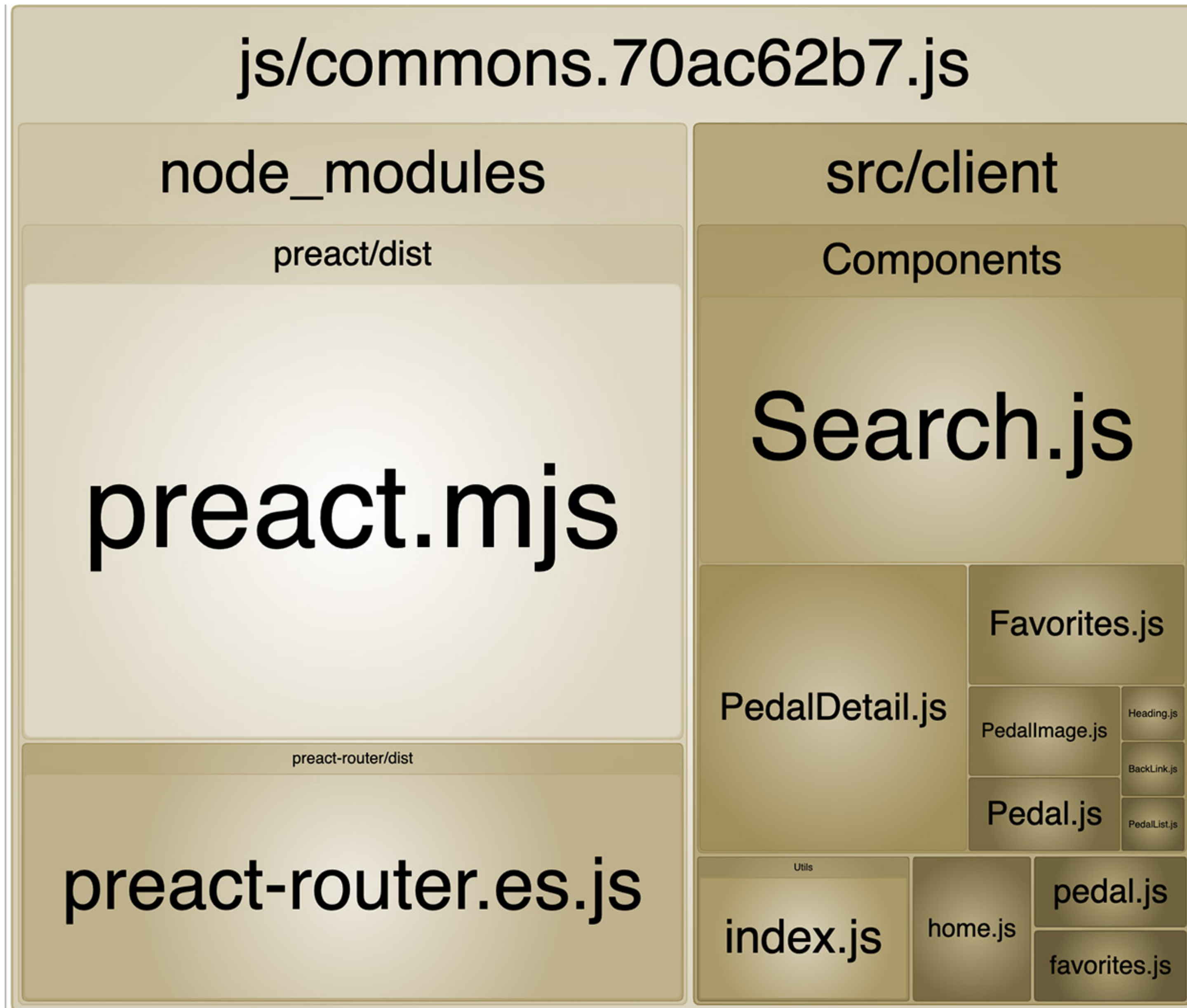
Search modules:

Enter regexp

x

Show chunks:

- ☒ All (26.75 KB)
- ☒ js/commons.70ac62b7.js (22.32 KB)
- ☒ js/favorites.8c4199f2.js (1.48 KB)
- ☒ js/pedal.b0a6b91b.js (1.47 KB)
- ☒ js/home.1350bbcf.js (1.47 KB)



js/favorites.8c4199f2.js

js/pedal.b0a6b91b.js

js/home.1350bbcf.js



MODERN BROWSERS: 26.75 KB



IE/Edge	Firefox	Chrome	Safari	fetches module	fetches nomodule	executes	
15-	59-	55-	10.0-	v	v	nomodule	✗
16			10.1/3	v	v	module	✗
17-18				double!	v	module	✗✗
		56-60			v	nomodule	✓
	60+	61+	11.0+*	v		module	✓

CREDIT: JAKUB GIERYLUK

[HTTPS://JAKUB.GIERYLUK.NET](https://jakub.gieryluk.net)

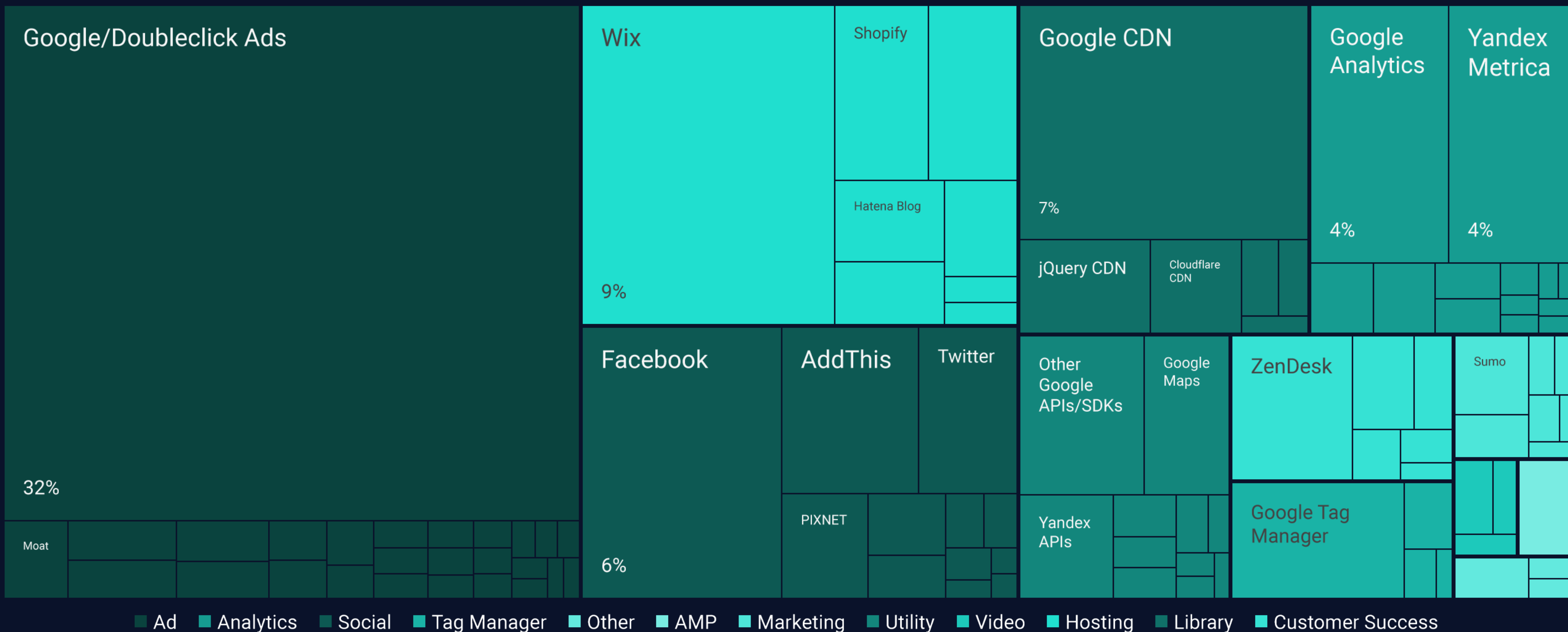

```
var scriptEl = document.createElement("script");

if ("noModule" in scriptEl) {
    scriptEl.src = "/js/scripts.mjs";
    scriptEl.type = "module";
} else {
    scriptEl.src = "/js/scripts.js";
    scriptEl.defer = true;
}

document.body.appendChild(scriptEl);
```

THIRD PARTY

CRASHER

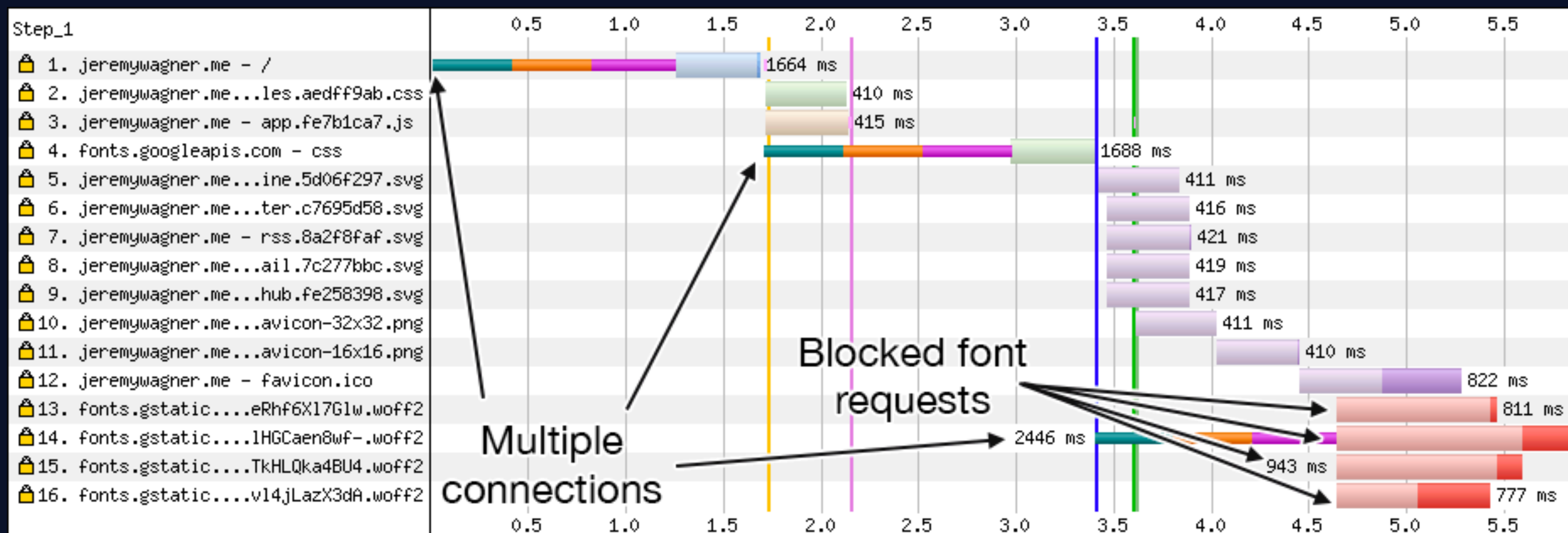


CREDIT: PATRICK HULCE

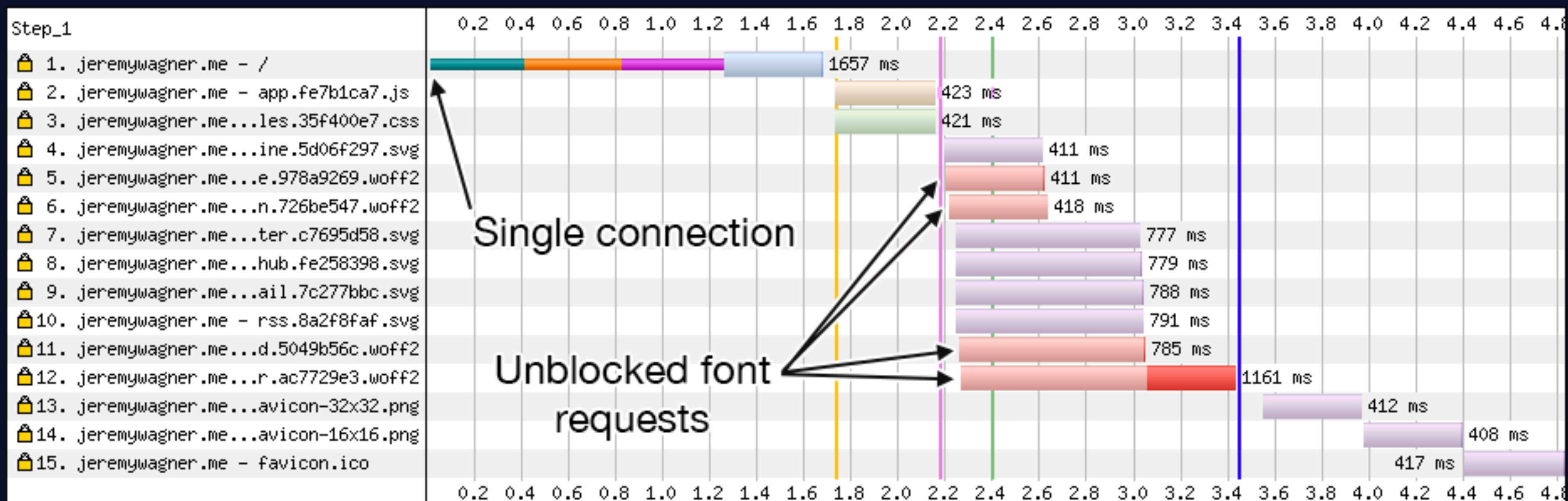
[HTTPS://WWW.THIRDPARTYWEB.TODAY](https://www.thirdpartyweb.today)

Index	URL	Resource Type	Size (KB)	Transfer Size (KB)	Time (ms)	Status
62.	gum.criteo.com - syncframe	Image	1982	1982	1982 ms	200
63.	www.cnn.com - sprite-chrome.png	Image	734	734	734 ms	200
64.	www.i.cdn.cnn....nsans-light.woff2	Font	998	998	998 ms	200
65.	logx.optimizely.com - events	Image	2188	2188	2188 ms	200
66.	amplifypixel.outbrain.com - pixel	Image	2403	2403	2403 ms	200
67.	data.cnn.com - domestic.json	Text	683	683	683 ms	200
68.	match.adsrvr.org - rid	Image	2114	2114	2114 ms	200
69.	mid.rkdms.com - ids	Image	2126	2126	2126 ms (404)	404
70.	api.rlcdn.com - identity	Image	2167	2167	2167 ms	200
71.	s.amazon-adsystem.com - iu3	Image	2415	2415	2415 ms (302)	302
72.	s.amazon-adsystem.com - iu3	Image	231	231	231 ms	200
73.	widgets.outbrain.com - outbrain.js	Script	1601	1601	1601 ms	200
74.	smetrics.cnn.com - s73799169356339	Image	939	939	939 ms (302)	302
75.	secure-us.imrworldwide.com - m	Image	946	946	946 ms (302)	302
76.	w.usabilla.com - 0649ef72a7be.js	Script	657	657	657 ms	200

⏮ ⏪ Elements Console Network Sources Performance Memory Application Security »					● 🚫 🔍 ☐ Preserve log ☐ Disable cache Online ▼				
-domain:*.cnn.com ☐ Hide data URLs All XHR JS CSS Img Media Font Doc WS Manifest Other									
Name		Protocol	Domain	Size	Time				
DC gpt.js		http/2+quic/43	www.googletagservices.com	10.7 KB	95 ms				
A9 apstag.js		http/1.1	c.amazon-adsystem.com	19.7 KB	247 ms				
CR publishertag.js		http/1.1	static.criteo.net	25.5 KB	343 ms				
131788053.js		h2	cdn.optimizely.com	105 KB	407 ms				
d22a6b50-6c73-4070-8922-86c722dd1f3...		h2	cdn.cookie law.org	992 B	179 ms				
obtp.js		http/1.1	amplify.outbrain.com	2.6 KB	259 ms				
JS freewheel-mapping.json		h2	cdn.jsdelivr.net	1.7 KB	95 ms				



FONTS HOSTED BY GOOGLE FONTS



SELF-HOSTED FONTS

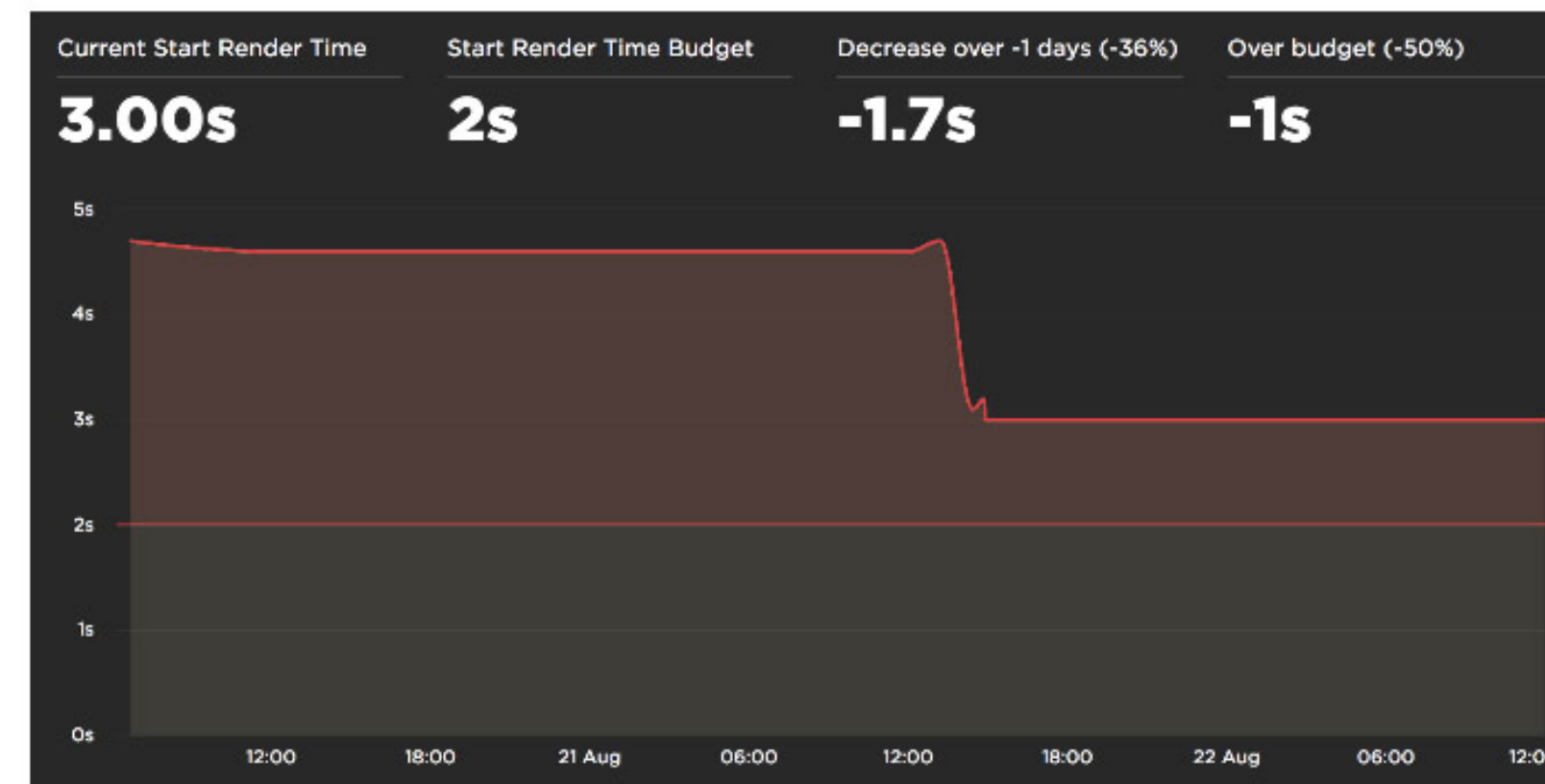
How we shaved 1.7 seconds off casper.com by self-hosting Optimizely



Kyle Rush

Aug 28, 2018 · 7 min read

We recently deployed a change to casper.com that loaded a piece of 3rd party JavaScript from our own server instead of the vendor's server. This change shaved 1.7 seconds off of the start render time:



Measurement taken on Desktop Chrome w/3G network connection

The 3rd party JavaScript in question is from a company called Optimizely. We use their client side JavaScript to conduct a/b tests on casper.com. Once the JavaScript file is downloaded and executed, it changes the document for 50% of our website visitors to measure how they react to the change. To ensure the smallest possible flash of unstyled content (FOUC) we follow the best practice of loading Optimizely before anything else, in a blocking manner.

As would be expected, loading the JavaScript snippet in this way has a negative impact on the web performance of our website. It's a trade-off we struggled with for a long time. Should we follow web performance best practices and load the Optimizely JavaScript asynchronously, or follow experimentation best practices and load it blocking as the first asset? Both of


```
<!-- Stick resource hints like this in your <head>: -->  
<link rel="dns-prefetch" href="https://fonts.googleapis.com/">  
<link rel="preconnect" href="https://fonts.googleapis.com/" crossorigin>
```

BE ACCOMMODATING

The Unacceptable Persistence of the Digital Divide

Millions of Americans lack broadband access and computer skills. Can President Trump bring them into the digital economy?

by David Talbot December 16, 2016

M

ost homes in the United States have Internet service, but they

don't in the poor parts of Cleveland and nearby suburbs. **A**

survey in 2012 showed that 58 percent of the area's

households with incomes under \$20,000 had neither home broadband

nor mobile Internet access, often because of the cost. Another 10

percent had a mobile phone but no home broadband. Until recently, one

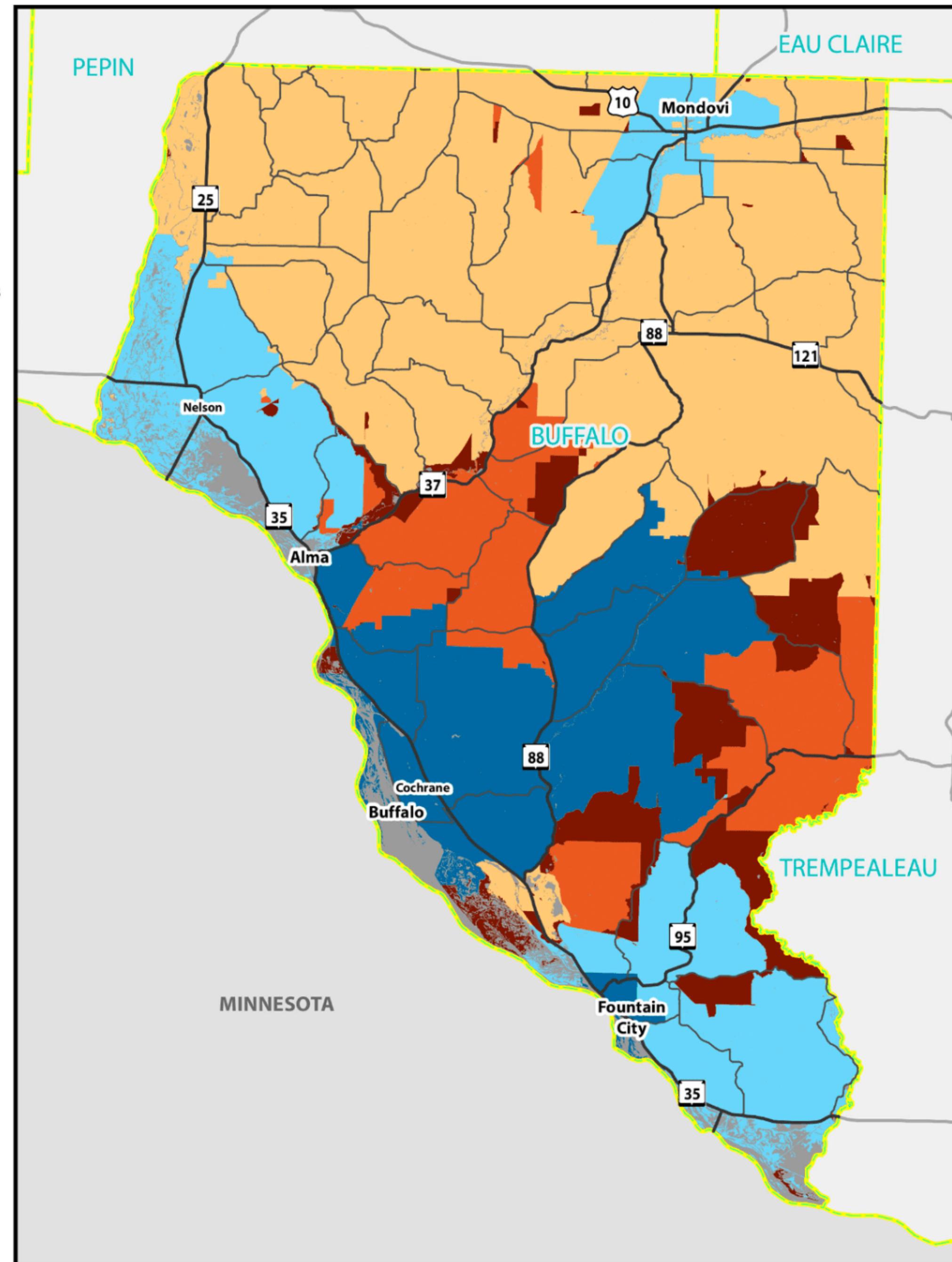
Marcella and Ma’Niyah are among the millions of people on the wrong side of America’s persistent digital divide. A survey by Pew Research shows that fully one-third of American adults do not subscribe to any Internet access faster than dial-up at their home at a time when many basic tasks—finding job listings, doing homework, obtaining social services, and even performing many jobs—require being online. Even many people who are willing to pay for service can’t get it. Thirty-four million Americans have no access at all to broadband as the U.S. Federal Communications Commission defines it: a download speed of at least 25 megabits per second and an upload speed of three megabits per second. These speeds are what FCC chairman Tom Wheeler calls “table stakes for 21st-century communications.”

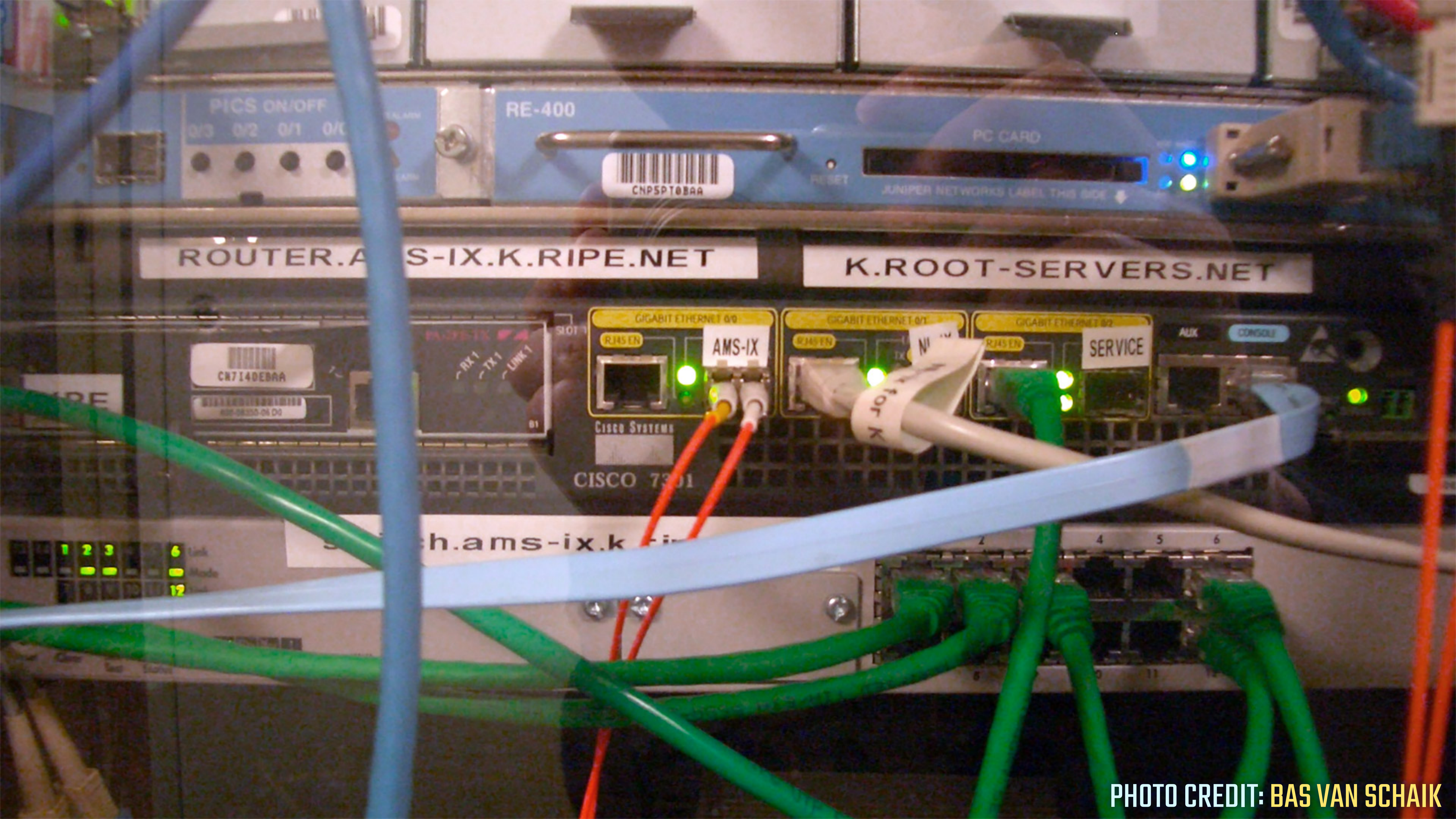
Buffalo County Internet Speeds

Includes Wireline and Fixed
Wireless Technologies
Download Speed / Upload Speed
(Megabits per Second)

- 25+ Mbps Down / 3+ Mbps Up
- 10 Mbps Down / 1.5 Mbps Up
- 3 Mbps Down / .768 Mbps Up
- Less than 3 / .768 Mbps
- Unserved for Fixed Internet Access
- Body of Water

0 1.25 2.5 5 Miles





PICS ON/OFF
0/3 0/2 0/1 0/0

RE-400

CNP5PT0BAA

PC CARD

RESET

JUNIPER NETWORKS LABEL THIS SIDE

ROUTER.AMS-IX.K.RIPE.NET

K.ROOT-SERVERS.NET

CN7140EBAA

800-08355-06 D0

GIGABIT ETHERNET 0/0
RJ45 EN
AMS-IX

GIGABIT ETHERNET 0/1
RJ45 EN
K.ROOT-SERVERS.NET

GIGABIT ETHERNET 0/2
RJ45 EN
SERVICE

CISCO SYSTEMS

CISCO 7301

h.ams-ix.k

PHOTO CREDIT: BAS VAN SCHAIK

RTT

Downlink

ECT

RTT

Approximate round trip time (*ms*)

Downlink

Approximate download speed (*kbps*)

ECT

Effective connection type ("4g", "3g", "2g", "*slow-2g*")

Accept-CH: RTT, Downlink, ECT

Accept-CH-Lifetime: 86400

```
<?php
$sect = "4g" ;

if (isset($_SERVER["HTTP_ECT"])) {
    $sect = $_SERVER["HTTP_ECT"] ;
}
?>
```



```
<?php
if ($sect === "4g" || $sect === "3g") {
    ?>
    <div class="carousel">
        <!-- Carousel content... -->
    </div>
    <script defer src="/js/carousel.js"></script>
    <?php
}
?>
```

ADAPTIVE PERFORMANCE

555-555-5555



INFO@SCONNIETIMBER.COM

SCONNIE
TIMBER

INC

OUR PROCESS

LOCATIONS

CONTACT US

Our management team believes we can benefit central Wisconsin's landowners best by providing additional services that compliment commercial logging operations. As industry leaders, we strive to continue to provide excellent service by employing two accredited foresters that can assist landowners with a variety of management objectives. See our Q&A section below to better understand the process and some of the details associated with working with us.



Q&A

► Q: How many years have you been in business?

► Q: How many timber sales do you do in a year?

► Q: How long does it take to complete a timber sale?

► Q: How long does it take from the beginning of the process to the

WISCONSIN'S LEADER IN FOREST
MANAGEMENT AND LOGGING
SINCE 1952

22 REQUESTS, 740 KB

91.26 SECONDS OVER 2G

555-555-5555



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SCONNIE
TIMBER

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Q&A

Q: How many years have you been in business?

A: We have been in business since 1952. We can't believe we have had the honor of working with central Wisconsin landowners for over 40 years. We are proud of our past but we are always looking forward to the future and to make the next 40 years even better.

Q: How many timber sales do you do in a year?

A: In a typical year we do around 175 individual timber sales. These sales are a mixture of state land, county land, MFL properties and private land.

Q: How long does it take to complete a timber sale?

A: The time frame can vary based on many factors. Acreage, type of trees to be harvested, type of harvest being conducted and ground/soil conditions can all affect the time frame of a harvest.

Q: How long does it take from the beginning of the process to the end?

A: The first step is to contact one of our foresters to schedule an appointment to look at your property. From there we will send a bid on the designated timber to be harvested, followed by a timber sale contract if everything looks good to you in the bid. Once we have a signed timber sale contract in house we will schedule any marking that needs to be done and schedule the

WISCONSIN'S LEADER IN FOREST
MANAGEMENT AND LOGGING
SINCE 1952

5 REQUESTS, 12 KB

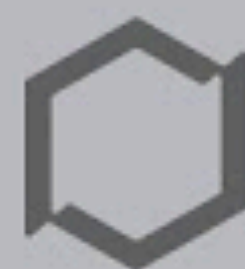
5.17 SECONDS OVER 2G



Width
The optimal width for
a requested image resource.

TAKE A (CLIENT) HINT!
JEREMY WAGNER
@MALCHATA

DIAMOND SPONSORS



BCG
Digital
Ventures



@FULLSTACKFEST 
#FULLSTACKFEST 
ORGANIZED BY :codegram

FULL
STACK
FEST



Web Fundamentals

GUIDES

CODE LABS

SAMPLES

[Replace Animated GIFs with Video](#)[JavaScript Start-up Optimization](#)[Loading Third-Party JavaScript](#)[Web Font Optimization](#)[HTTP Caching](#)[Adapting to Users with Client Hints](#)[Delivering Fast and Light Applications with Save-Data](#)[▶ Optimizing JavaScript](#)[▶ Never Load the Same Resource Twice](#)[▶ Lazy Loading Resources](#)[▶ Order Loading Thoughtfully](#)[PRPL Pattern](#)[Resource Prioritization](#)[▶ Web Performance Optimization with webpack](#)[▶ Rendering Performance](#)[▶ Audit your site](#)

Adapting to Users with Client Hints

**By** [Jeremy Wagner](#)

Jeremy is a contributor to WebFundamentals

Developing sites that are fast everywhere can be a tricky prospect. The plethora of device capabilities—and the quality of the networks they connect to—can make it seem like an insurmountable task. While we can take advantage of browser features to improve loading performance, how do we know what the user's device is capable of, or the quality of their network connection? The solution is [client hints](#)!

Client hints are a set of opt-in HTTP request headers that give us insight into these aspects of the user's device and the network they're connected to. By tapping into this information server side, we can change *how* we deliver content based on device and/or network conditions. This can help us to create more inclusive user experiences.

It's All About Content Negotiation

Contents

[It's All About Content Negotiation](#)[Opting in](#)[All the client hints!](#)[Device hints](#)[Network hints](#)[Tying it all together](#)[Responsive Images](#)[Helping users on slow networks](#)[Mind those caches!](#)[Client hints in service workers](#)[Wrapping up](#)[Resources](#)

FIGURE OUT WHAT PEOPLE WANT
AND WORK BACKWARD FROM THERE





react-dom@16.8.6

 React package for working with the DOM

 4 dependencies

 npm



BUNDLE SIZE

103.7kB

MINIFIED

32.9kB

MINIFIED + GZIPPED

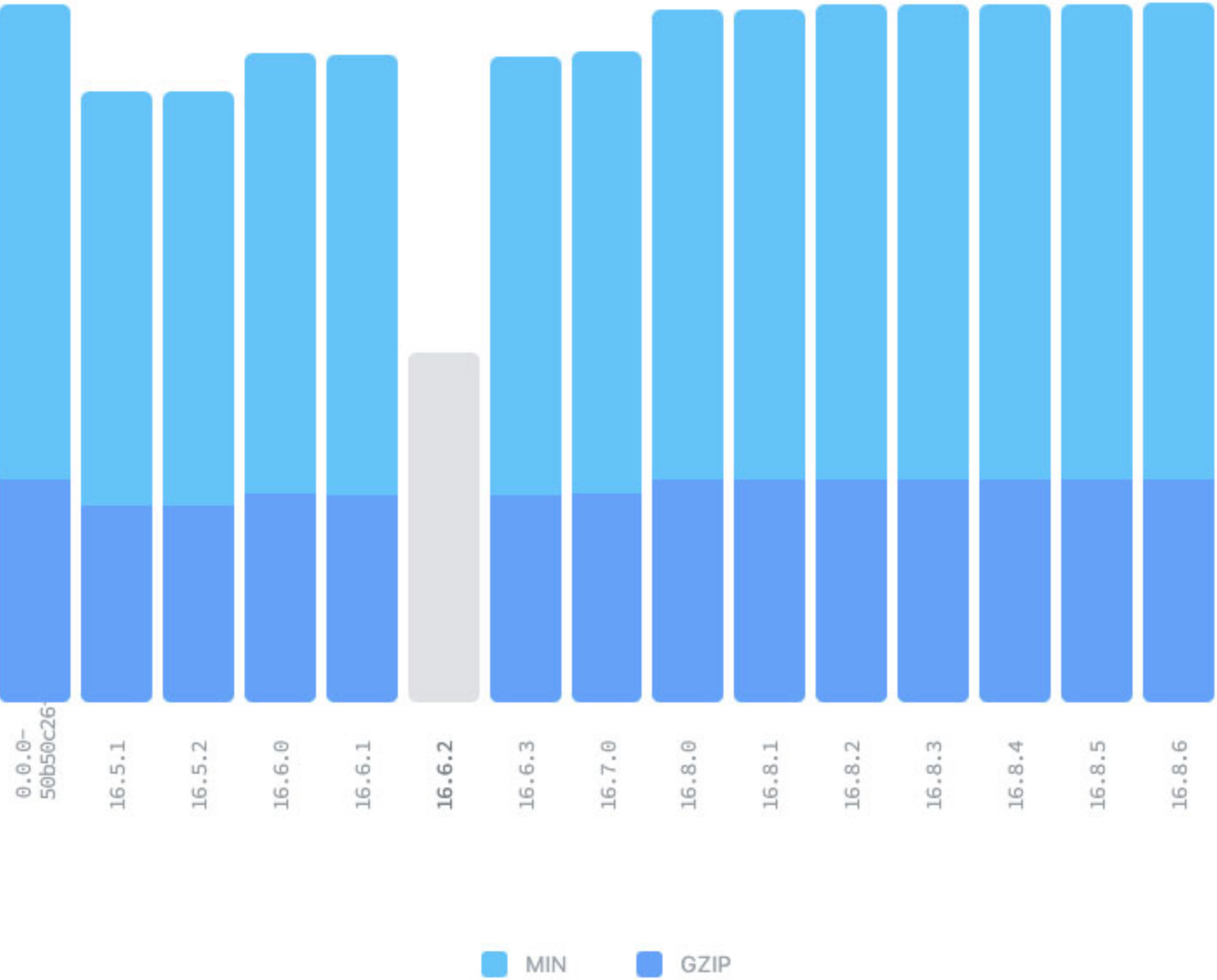
DOWNLOAD TIME

1.1s

2G EDGE ⓘ

0.66s

EMERGING 3G ⓘ



Composition



preact@8.4.2 

 Fast 3kb React alternative with the s...

 tree-shakeable

 no dependencies

 npm



BUNDLE SIZE

8.9kB

MINIFIED

3.5kB

MINIFIED + GZIPPED

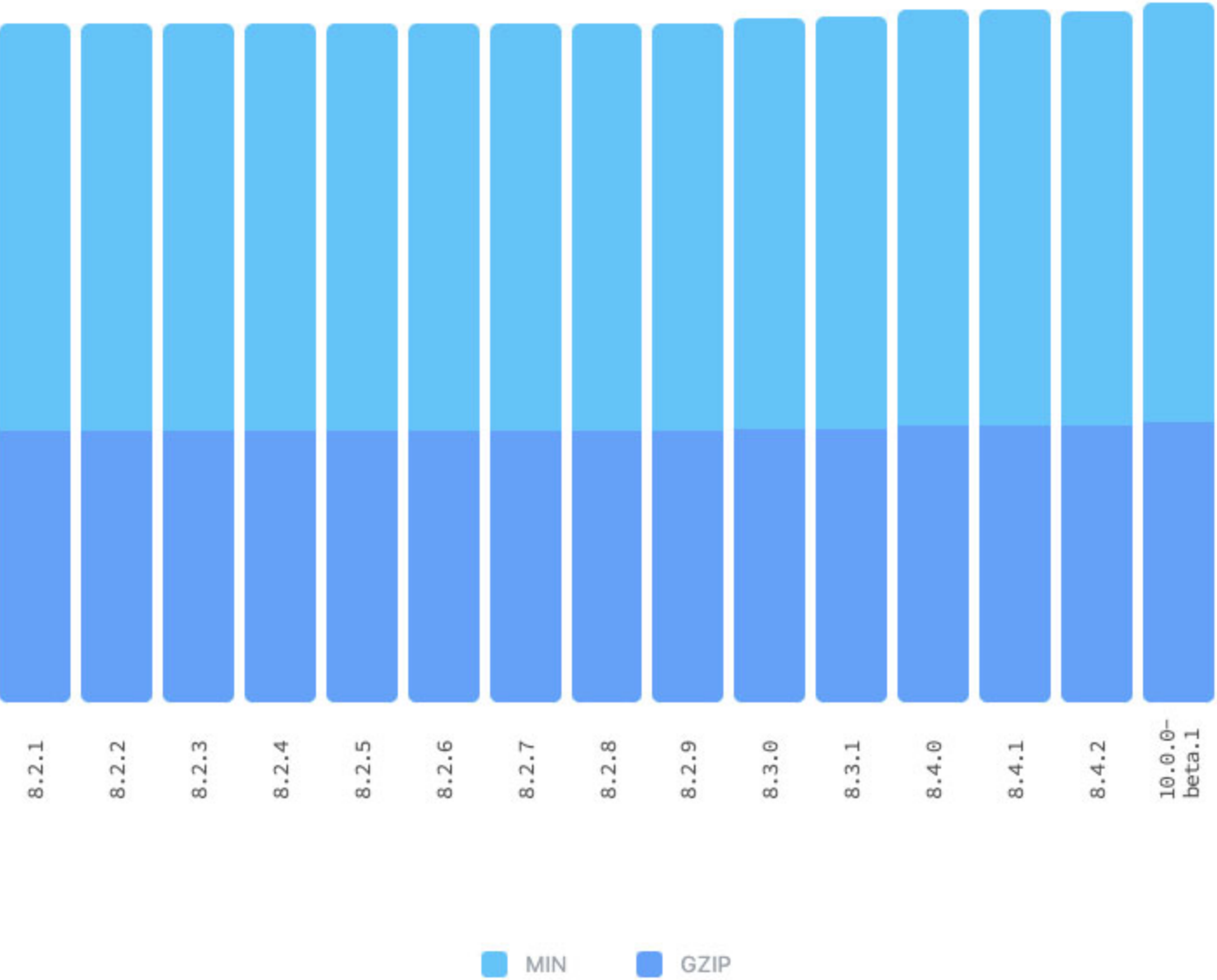
DOWNLOAD TIME

118ms

2G EDGE ⓘ

71ms

EMERGING 3G ⓘ



Composition



moment@2.24.0 

 Parse, validate, manipulate, and disp...

 tree-shakeable

 no dependencies

 npm



BUNDLE SIZE

231.7kB

MINIFIED

65.9kB

MINIFIED + GZIPPED

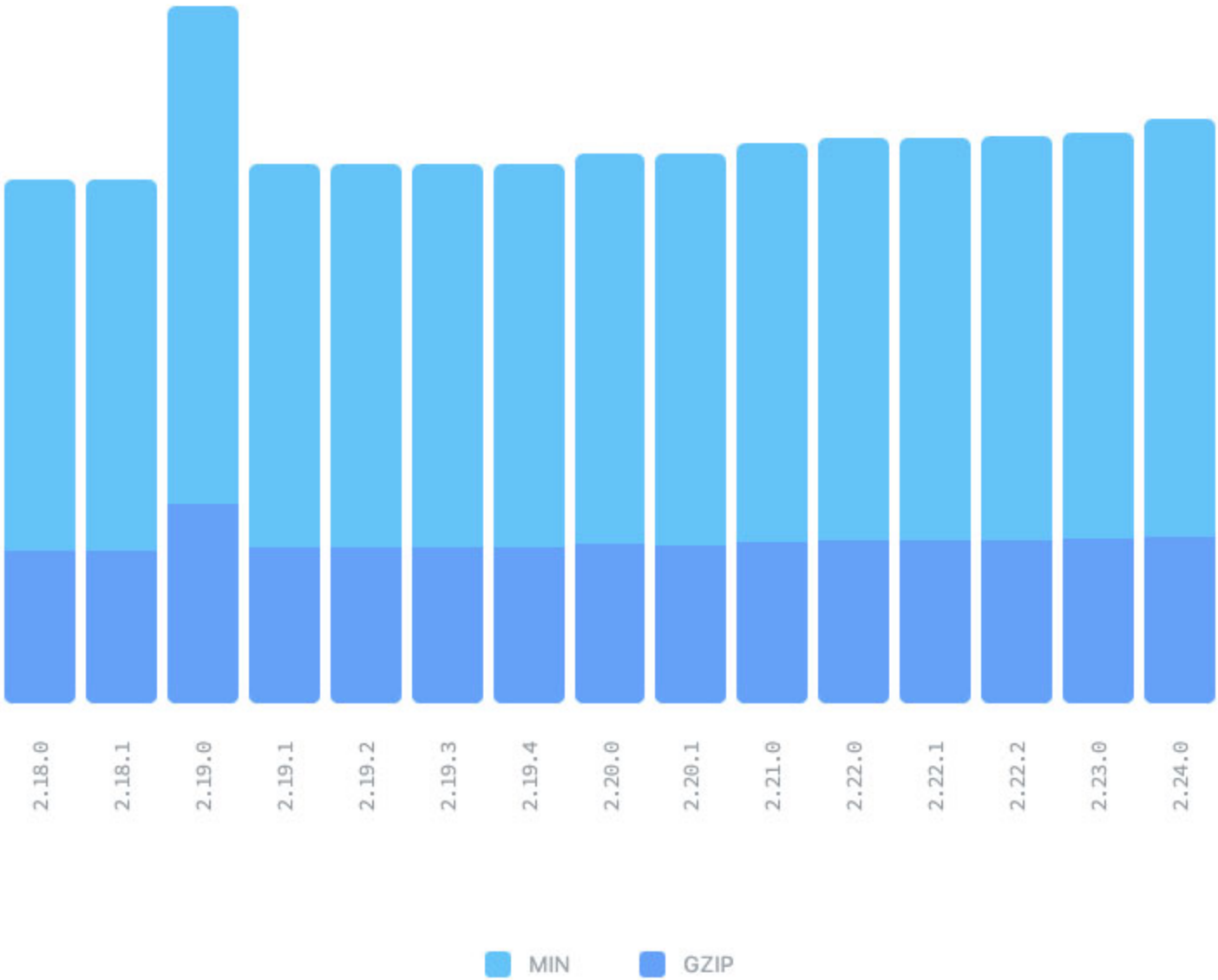
DOWNLOAD TIME

2.2s

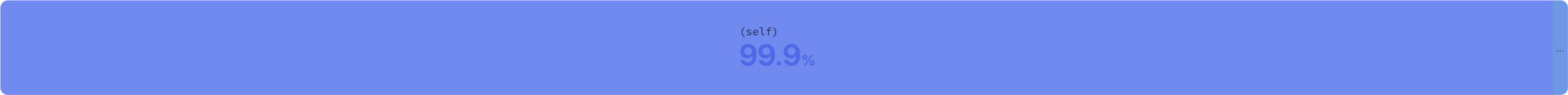
2G EDGE 

1.32s

EMERGING 3G 



Composition



date-fns@1.30.1 

 Modern JavaScript date utility library

 no dependencies

BUNDLE SIZE

29.3kB

MINIFIED

6.5kB

MINIFIED + GZIPPED

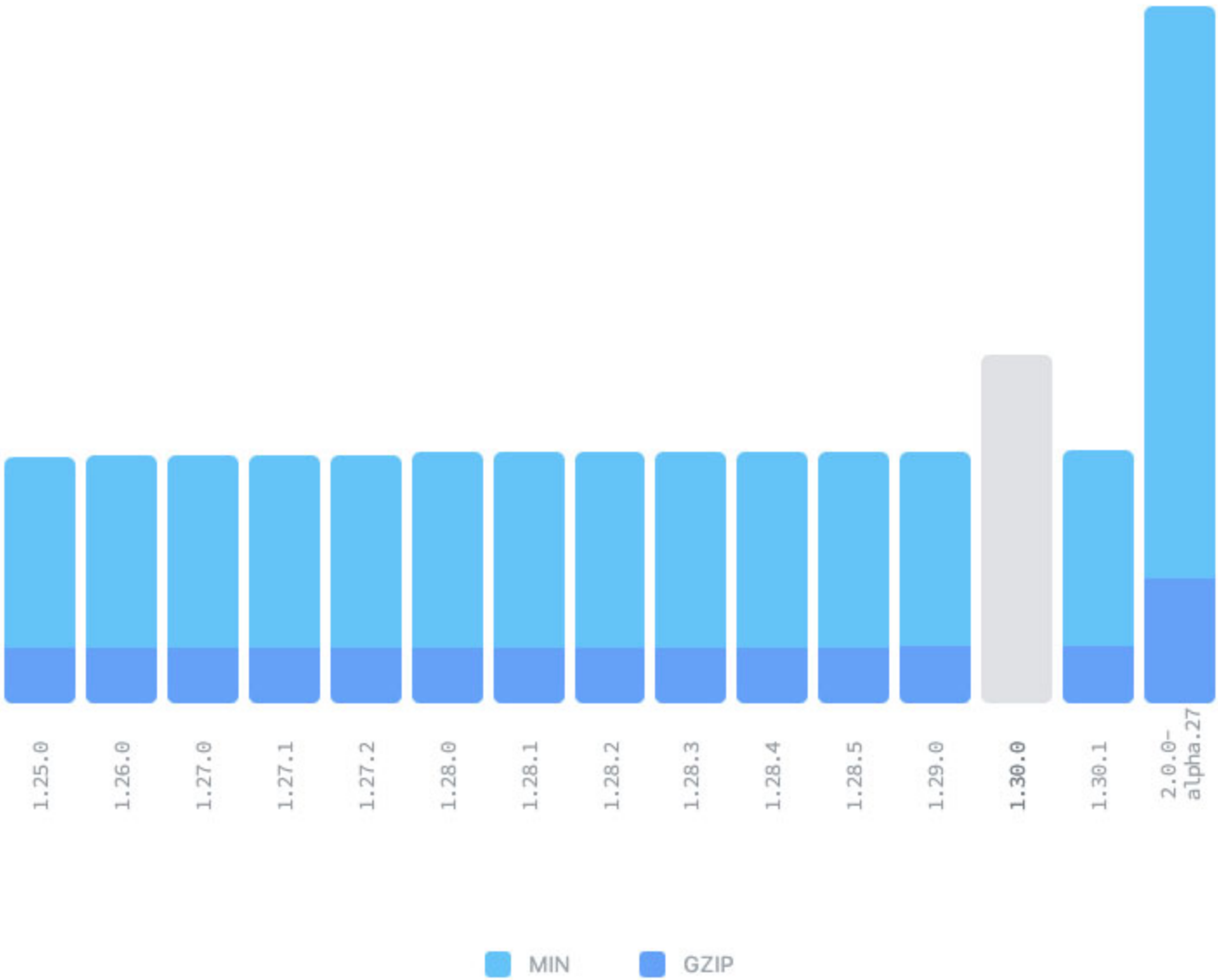
DOWNLOAD TIME

218ms

2G EDGE 

131ms

EMERGING 3G 



Composition



THANK YOU

JEREMY WAGNER — @MALCHATA — JEREMY.CODES

RESOURCES

RESPONSIBLE JAVASCRIPT: PART I

alistapart.com/article/responsible-javascript-part-1/

BUNDLEPHOBIA

bundlephobia.com

HTTP ARCHIVE

httparchive.org

WEBPAGETEST

webpagetest.org

PAINT THE PICTURE, NOT THE FRAME: HOW BROWSERS PROVIDE EVERYTHING USERS NEED

alistapart.com/article/paint-the-picture-not-the-frame

QUICKLINK

github.com/GoogleChromeLabs/quicklink

DNSTRADAMUS

github.com/malchata/dnstradamus

HYBRID LAZY LOADING: A PROGRESSIVE MIGRATION TO NATIVE LAZY LOADING

smashingmagazine.com/2019/05/hybrid-lazy-loading-progressive-migration-native

BABEL 6: LOOSE MODE

2ality.com/2015/12/babel6-loose-mode.html

THIRD-PARTY WEB

thirdpartyweb.today

HOW WE SHAVED 1.7 SECONDS OFF CASPER.COM BY SELF-HOSTING OPTIMIZELY

medium.com/caspertechteam/we-shaved-1-7-seconds-off-casper-com-by-self-hosting-optimizely-2704bcbff8ec

THE UNACCEPTABLE PERSISTENCE OF THE DIGITAL DIVIDE

technologyreview.com/s/603083/the-unacceptable-persistence-of-the-digital-divide

TAKE A (CLIENT) HINT!

youtube.com/watch?v=md7Ua82fPe4

ADAPTING TO USERS WITH CLIENT HINTS

developers.google.com/web/fundamentals/performance/optimizing-content-efficiency/client-hints